

4 Efficient Sorting

Efficient Algorithms · Wintersemester 2026–27

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CS566 (Wintersemester 2026–27)
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Learning Outcomes

Unit 4: ***Efficient Sorting***

1. Know principles and implementation of *mergesort* and *quicksort*.
2. Know properties and *performance characteristics* of mergesort and quicksort.
3. Know the comparison model and understand the corresponding *lower bound*.
4. Understand *counting sort* and how it circumvents the comparison lower bound.
5. Know ways how to exploit *presorted* inputs.

Outline

4 Efficient Sorting

4.1 Mergesort
4.2 Quicksort
4.3 Comparison-Based Lower Bound
4.4 Integer Sorting
4.5 Adaptive Sorting
4.6 Python's list sort

Why study sorting?

- fundamental problem of computer science that is still not solved
- building brick of many more advanced algorithms
  - for preprocessing
  - as subroutine
- playground of manageable complexity
  to practice algorithmic techniques

Algorithm with optimal #comparisons in worst case?

Here:

- "classic" fast sorting method
- exploit **partially sorted** inputs

Part I

The Basics

Rules of the game

- **Given:**
  - array $A[0..n) = A[0..n-1]$ of n objects
  - a total order relation $\leq$ among $A[0], \ldots, A[n-1]$
    (a comparison function)
    Python: elements support <= operator (`__le__()`)
    Java: `Comparable` class (`x.compareTo(y) <= 0`)
- **Goal:** rearrange (i. e., permute) elements within A,
  so that A is *sorted*, i. e., $A[0] \leq A[1] \leq \cdots \leq A[n-1]$

- for now: A stored in main memory (*internal sorting*)
  single processor (*sequential sorting*)

Clicker Question



What is the complexity of sorting? Type you answer, e. g., as "`Theta(sqrt(n))`"

lower bound: cf. $\Omega(n)$ time for subarray sum
upper bound (algorithm's Θ-class of running time)

$\Theta(n \log n)$



→ *sli.do/effalg*

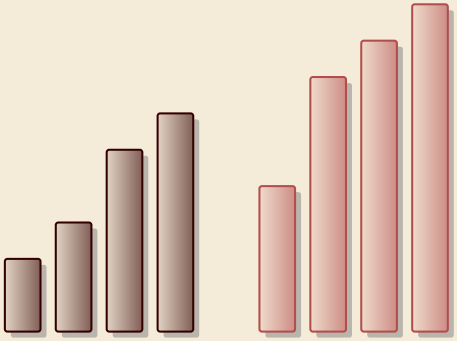
4.1 Mergesort

Merging sorted lists



Merging sorted lists

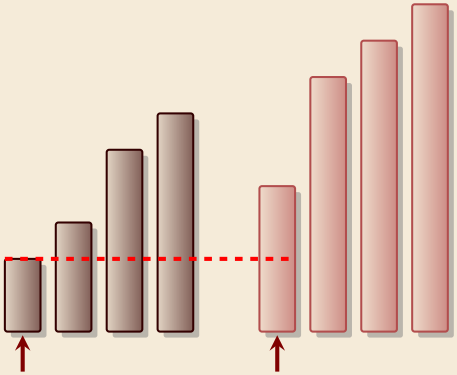




run1 run2 result

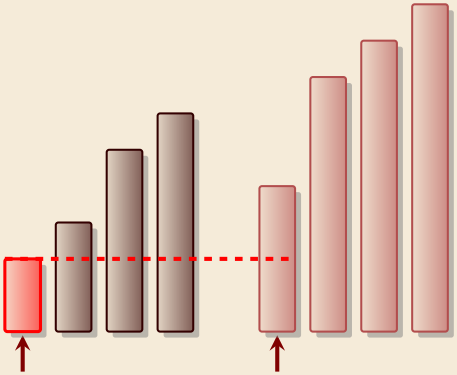
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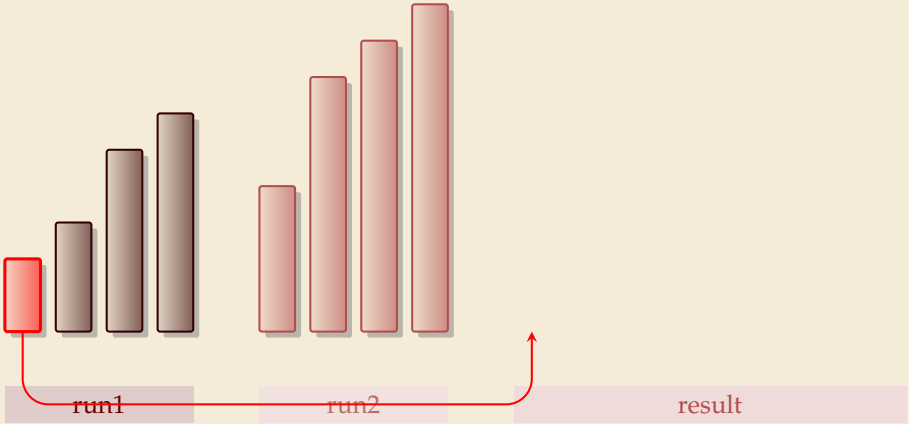
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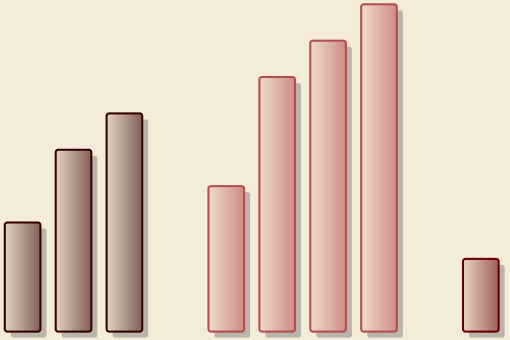
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Merging sorted lists





run1 run2 result

Merging sorted lists





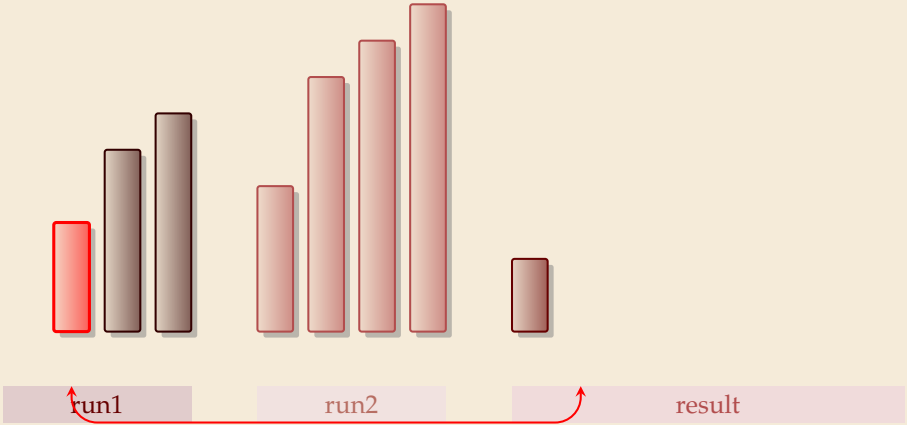
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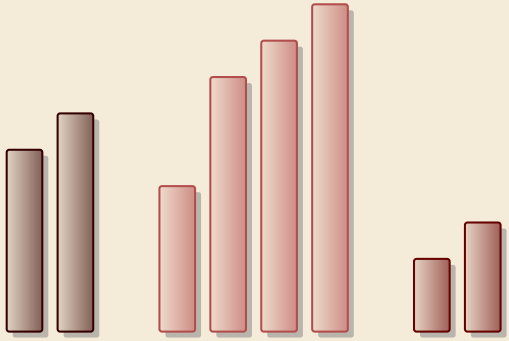
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Merging sorted lists





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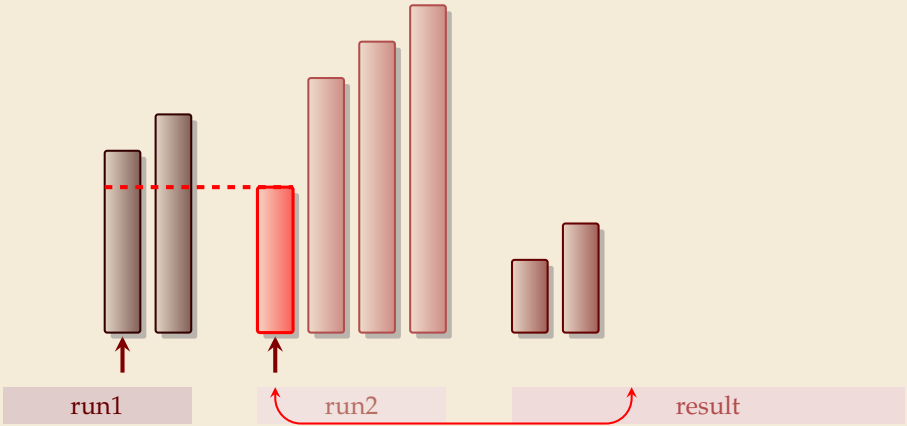
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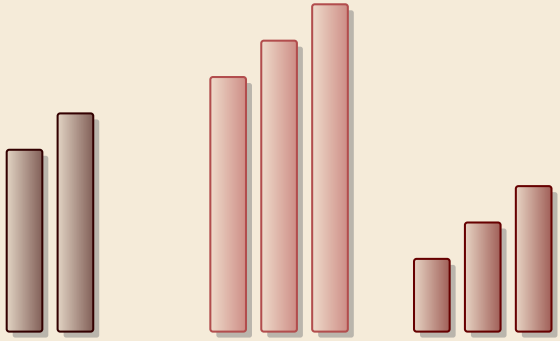
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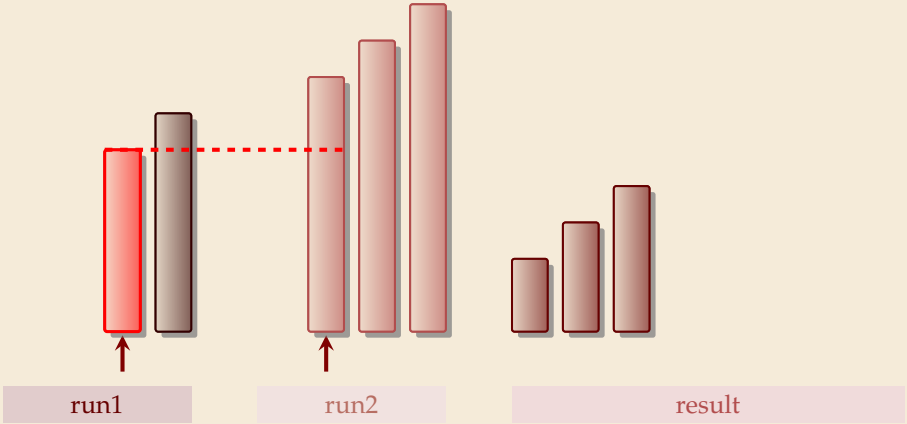




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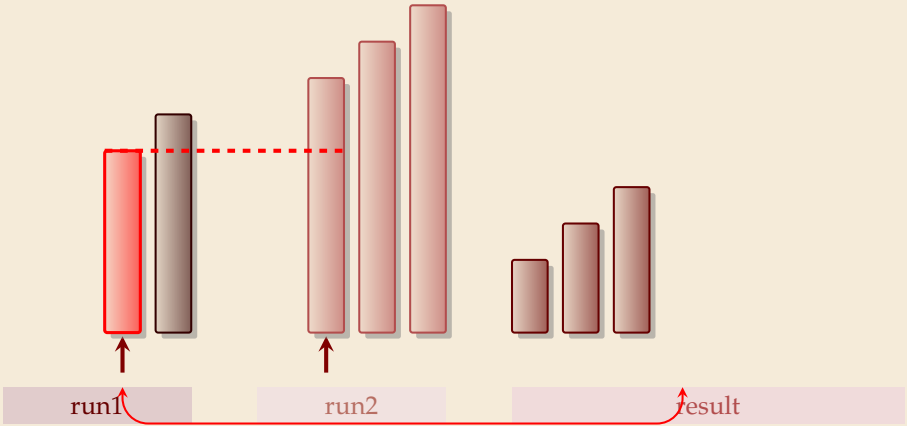
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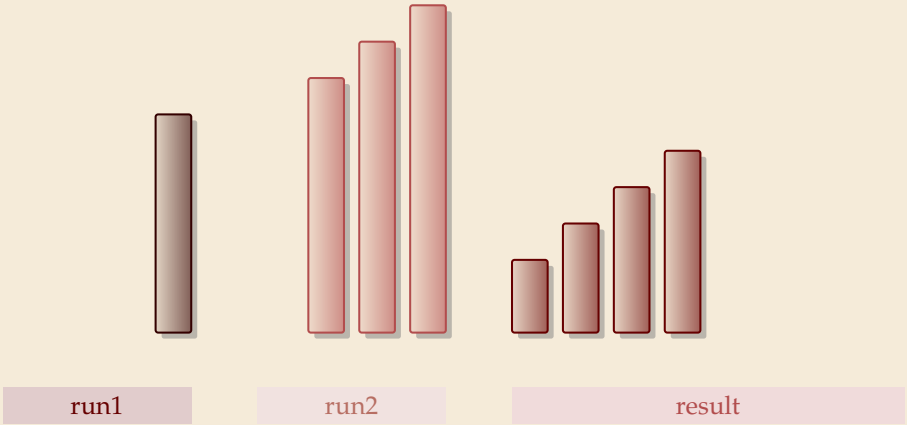
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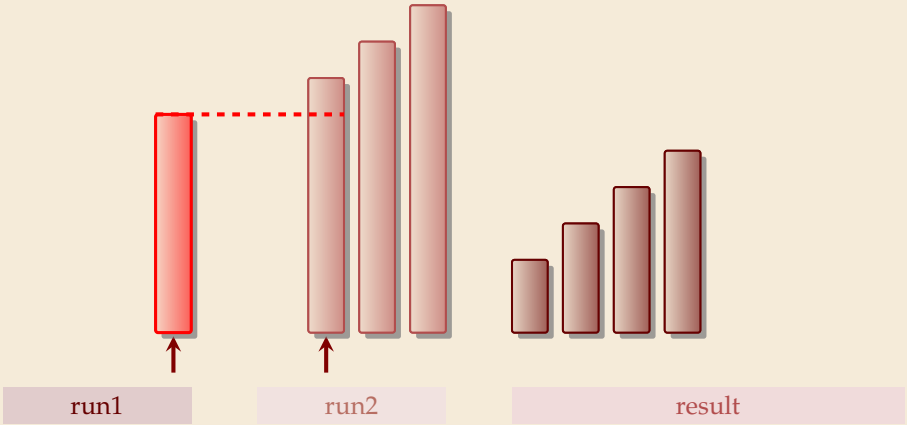
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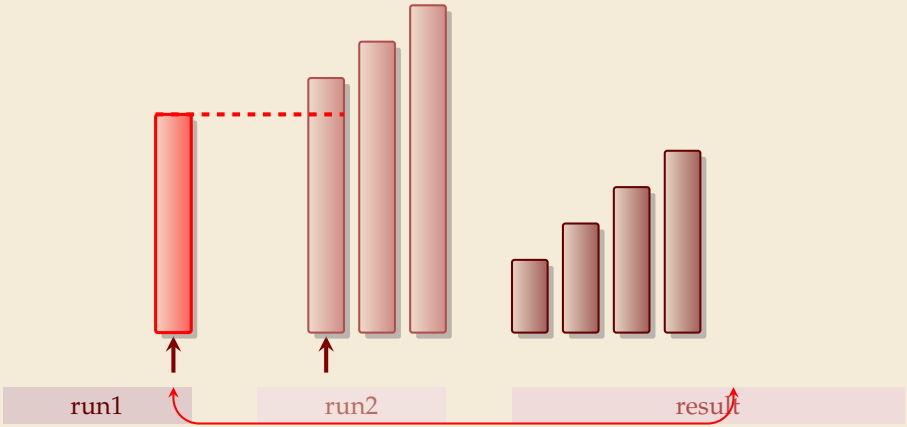
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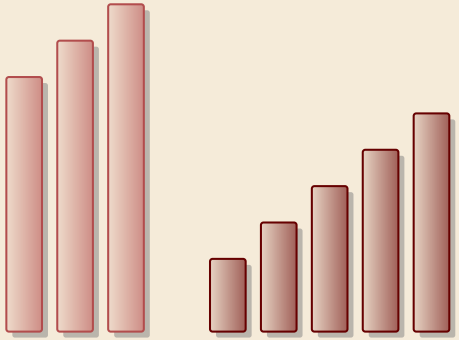
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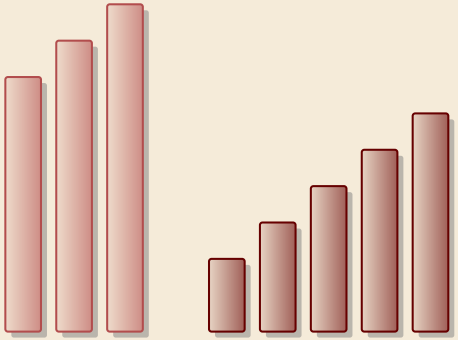




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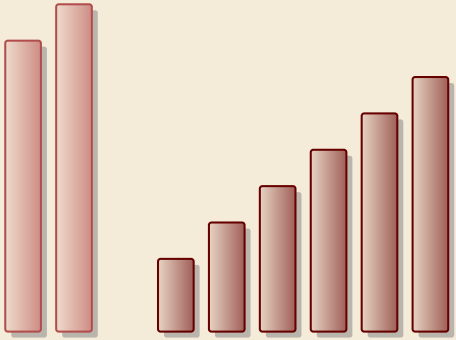




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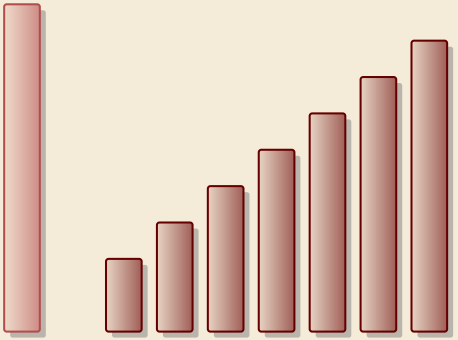




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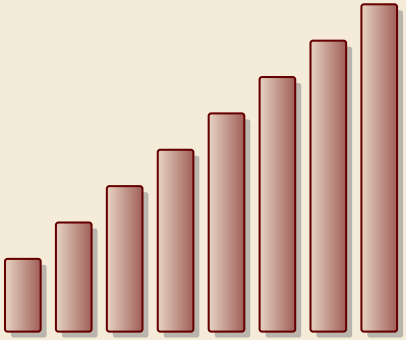




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Merging sorted lists





run1 run2 result

Clicker Question

What is the worst-case running time of mergesort?



- **A** $\Theta(1)$
- **B** $\Theta(\log n)$
- **C** $\Theta(\log \log n)$
- **D** $\Theta(\sqrt{n})$
- **E** $\Theta(n)$
- **F** $\Theta(n \log \log n)$
- **G** $\Theta(n \log n)$
- **H** $\Theta(n \log^2 n)$
- **I** $\Theta(n^{1+\epsilon})$
- **J** $\Theta(n^2)$
- **K** $\Theta(n^3)$
- **L** $\Theta(2^n)$



→ *sli.do/effalg*

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A	~~$\Theta(1)$~~	G	$\Theta(n \log n)$ ✓
B	~~$\Theta(\log n)$~~	H	~~$\Theta(n \log^2 n)$~~
C	~~$\Theta(\log \log n)$~~	I	~~$\Theta(n^{1+\epsilon})$~~
D	~~$\Theta(\sqrt{n})$~~	J	~~$\Theta(n^2)$~~
E	~~$\Theta(n)$~~	K	~~$\Theta(n^3)$~~
F	~~$\Theta(n \log \log n)$~~	L	~~$\Theta(2^n)$~~



→ *sli.do/effalg*

Mergesort

mergesort(A[0..n))

```
1 procedure mergesort(A[l..r)):
2     n := r − l
3     if n ≤ 1 return
4     m := l + ⌊n/2⌋
5     mergesort(A[l..m))
6     mergesort(A[m..r))
7     merge(A[l..m), A[m..r), buf)
8     copy buf to A[l..r)
```

- recursive procedure
- merging needs
  - temporary storage *buf* for result (of same size as merged runs)
  - to read and write each element twice (once for merging, once for copying back)

Mergesort

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  - to read and write each element twice (once for merging, once for copying back)

Analysis: count *"element visits"* (read and/or write)

$$C(n) = \begin{cases} 0 & n \le 1 \\ C(\lfloor n/2 \rfloor) + C(\lceil n/2 \rceil) + 2n & n \ge 2 \end{cases}$$

same for best and worst case!

Simplification $n = 2^k$

$$C(2^k) = \begin{cases} 0 & k \le 0 \\ 2 \cdot C(2^{k-1}) + 2 \cdot 2^k & k \ge 1 \end{cases} = 2 \cdot 2^k + 2^2 \cdot 2^{k-1} + 2^3 \cdot 2^{k-2} + \cdots + 2^k \cdot 2^1 = 2k \cdot 2^k$$

2^{k+1} — 1 — k copies

$$C(n) = 2n \lg(n) = \Theta(n \log n)$$

(arbitrary n: $C(n) \le C(\text{next larger power of 2}) \le 4n \lg(n) + 4n = \Theta(n \log n)$)

Mergesort

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same for best and worst case!

precisely(!) solvable *without* assumption $n = 2^k$:
$C(n) = 2n \lg(n) + \left(2 - \{\lg(n)\} - 2^{1-\{\lg(n)\}}\right)2n$
with $\{x\} := x - \lfloor x \rfloor$

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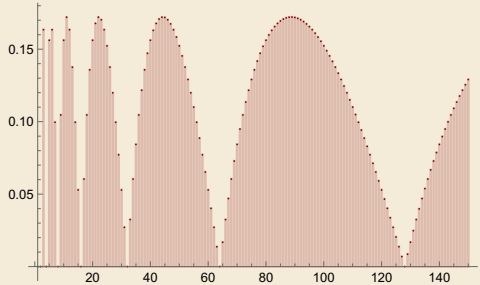
$C(n) = 2n \lg(n) = \Theta(n \log n)$ (arbitrary n: $C(n) \le C(\text{next larger power of 2}) \le 4n \lg(n) + 4n = \Theta(n \log n)$)

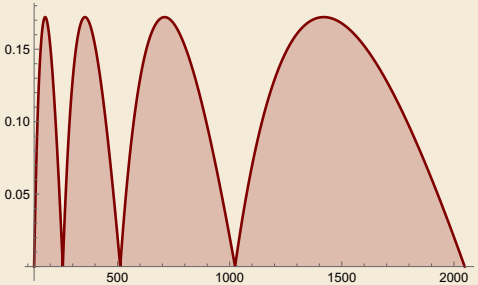
Linear Term of $C(n)$

Recall:

$$C(n) \;=\; 2n\lg(n) + \big(2 - \{\lg(n)\} - 2^{1-\{\lg(n)\}}\big)2n \qquad \text{with } \{x\} \coloneqq x - \lfloor x \rfloor$$

Plot of $2\big(2 - \{\lg(n)\} - 2^{1-\{\lg(n)\}}\big)$





$\rightsquigarrow$ Can prove: $C(n) \;\le\; 2n\lg n + 0.172n$

Mergesort – Discussion



optimal time complexity of $\Theta(n \log n)$ in the worst case



stable sorting method i. e., retains relative order of equal-key items



memory access is sequential (scans over arrays)



requires $\Theta(n)$ extra space

there are in-place merging methods,
but they are substantially more complicated
and not (widely) used

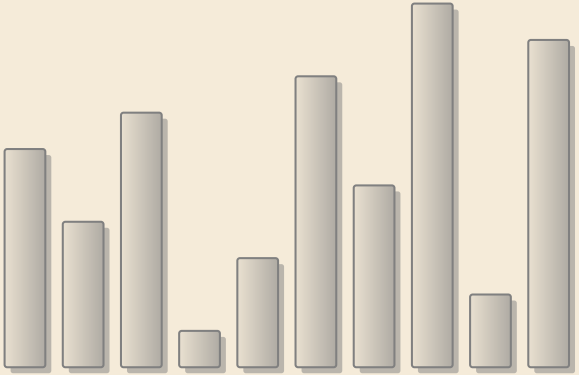
4.2 Quicksort

Partitioning around a pivot



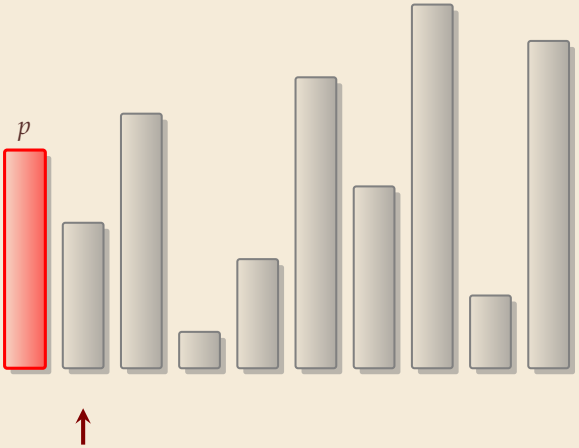
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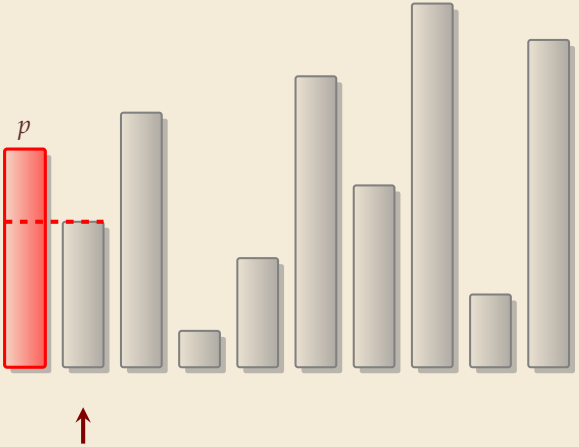
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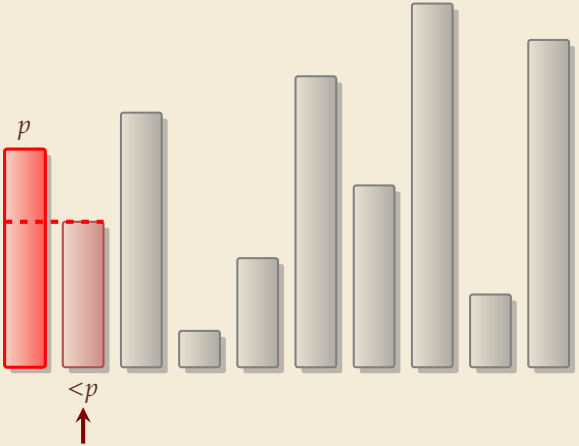
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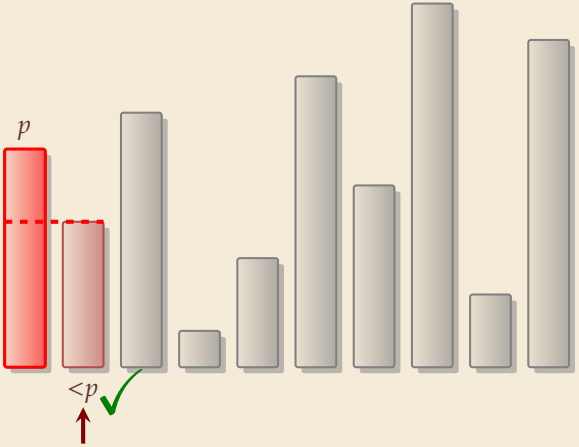
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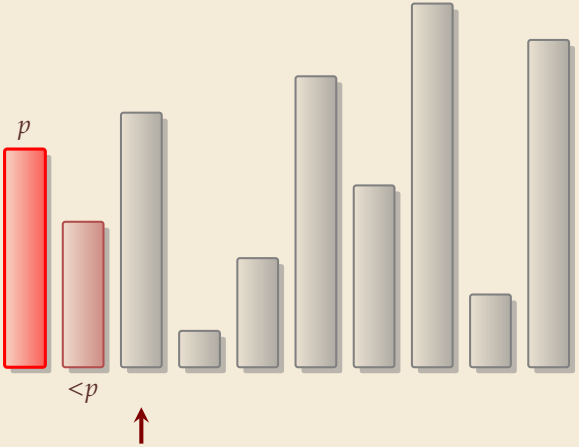
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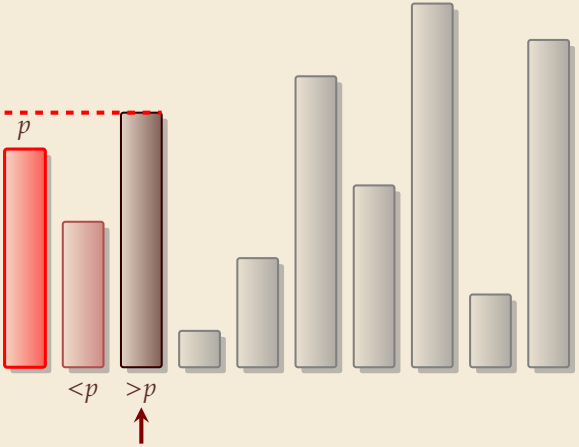
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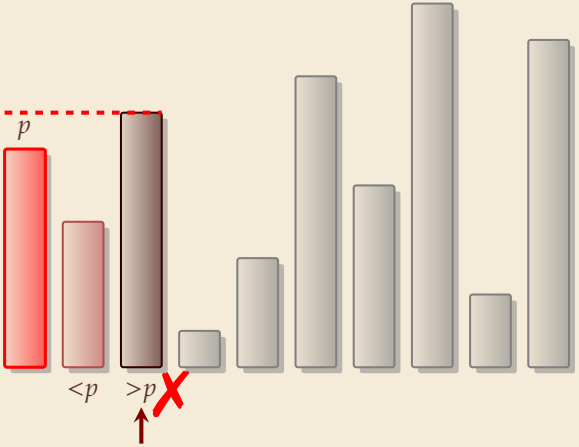
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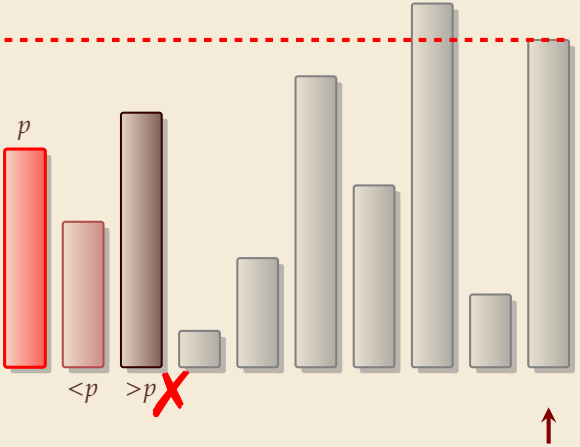
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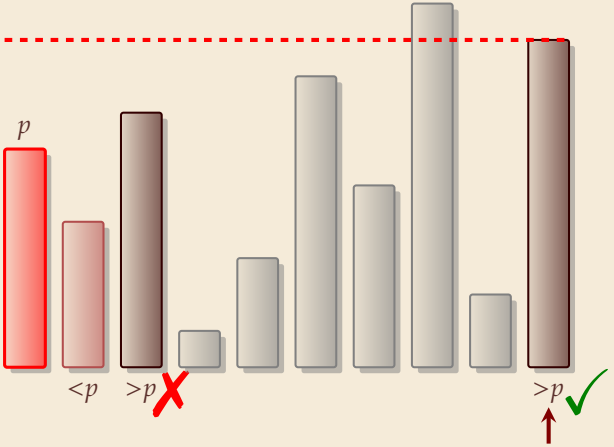
Partitioning around a pivot





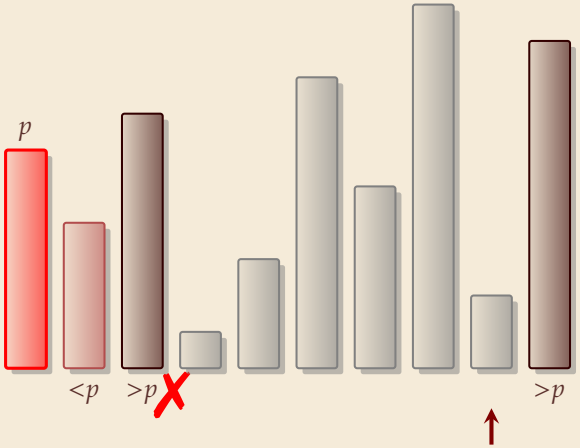
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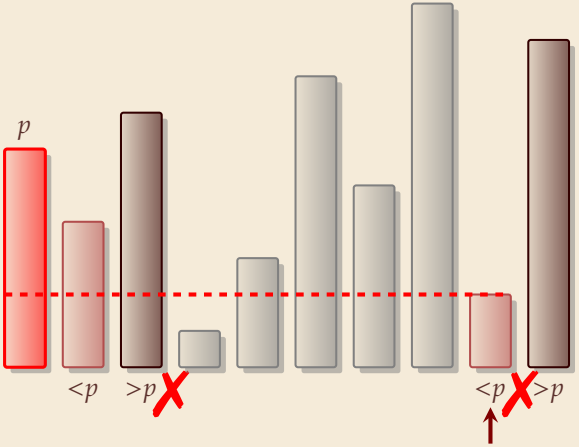
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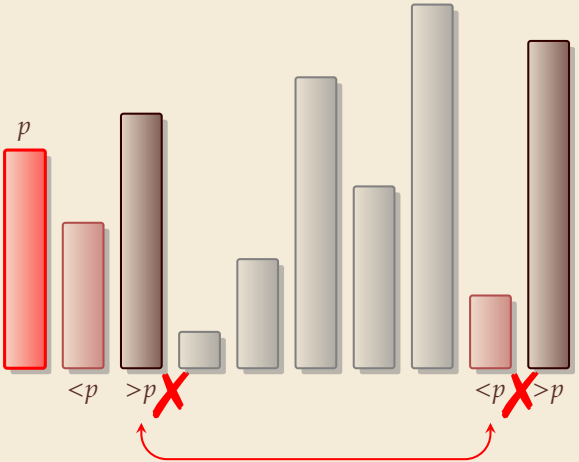
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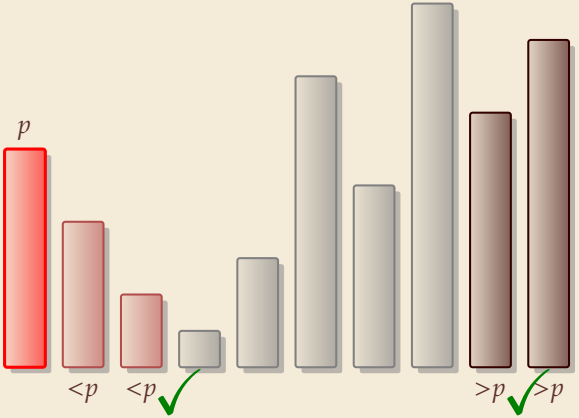
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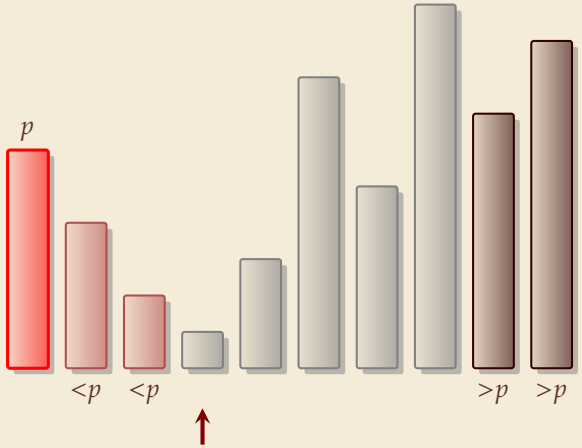
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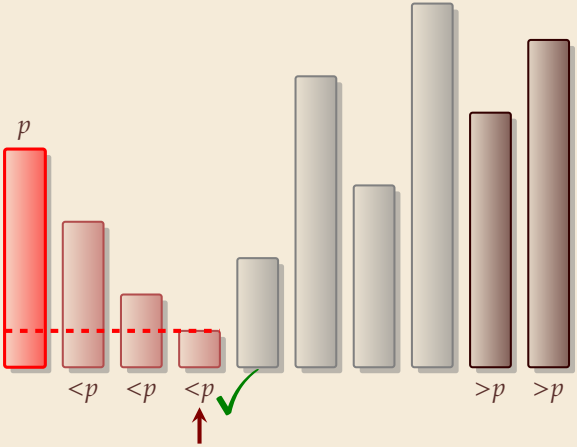
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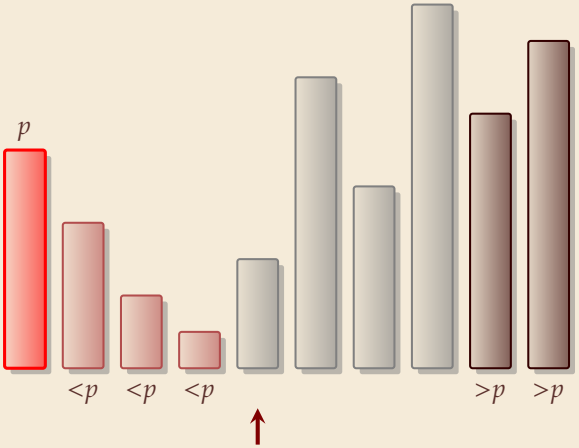
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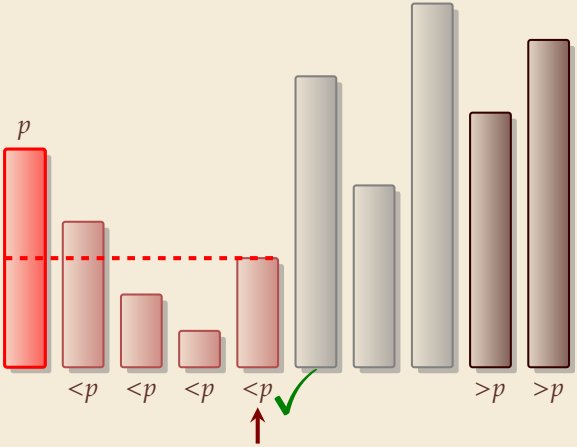
Partitioning around a pivot





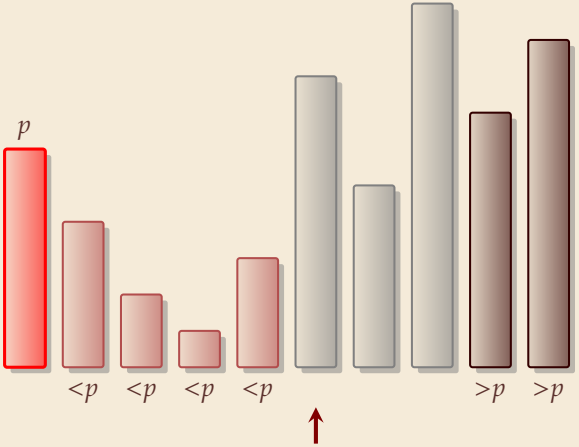
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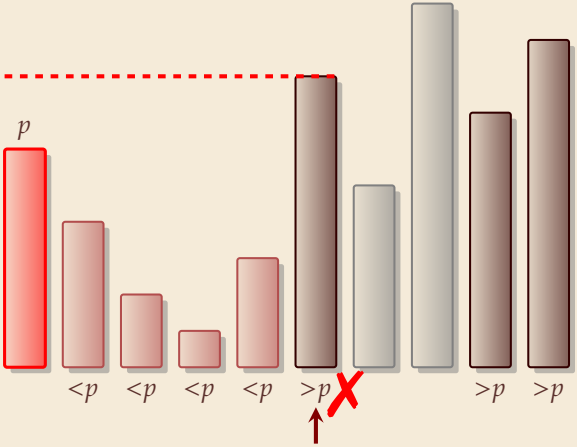
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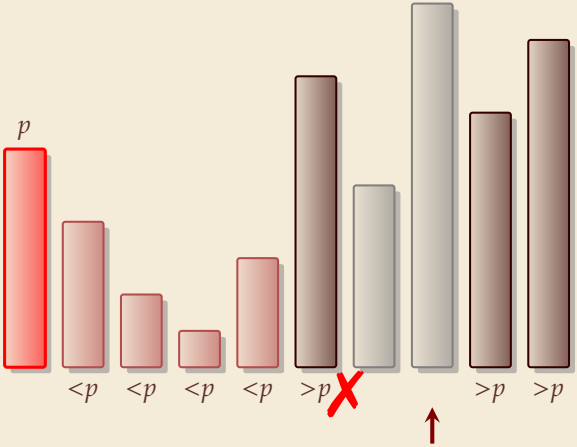
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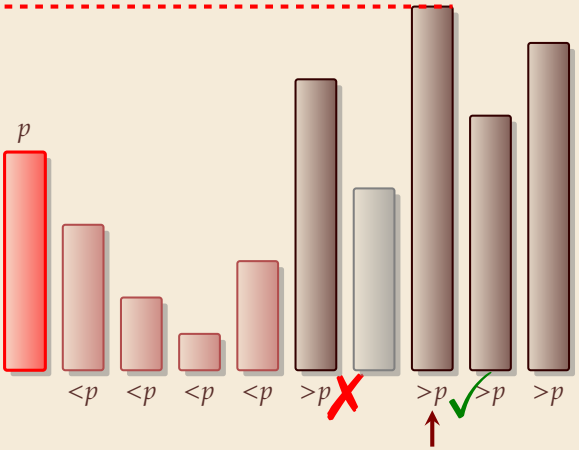
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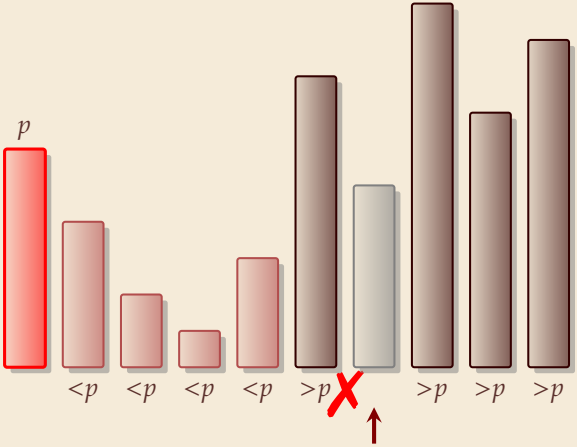
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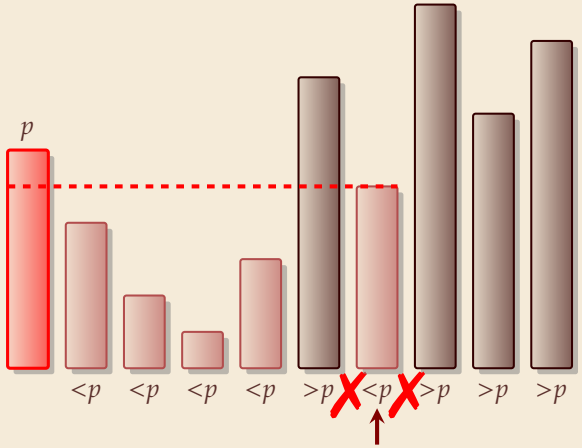
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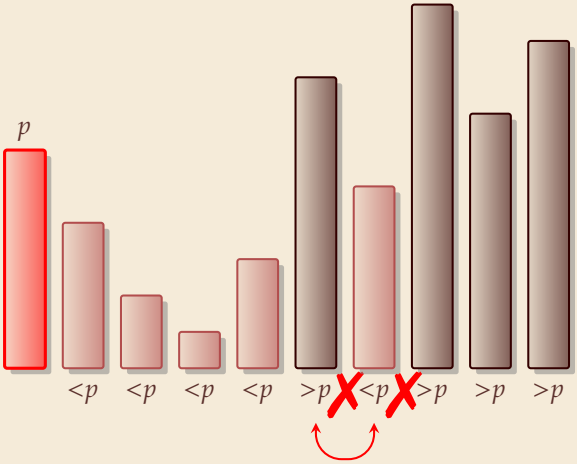
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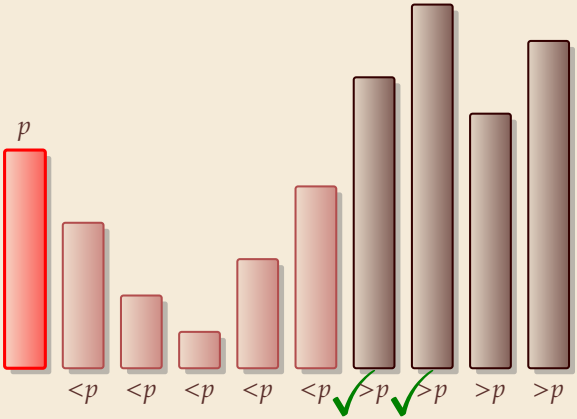
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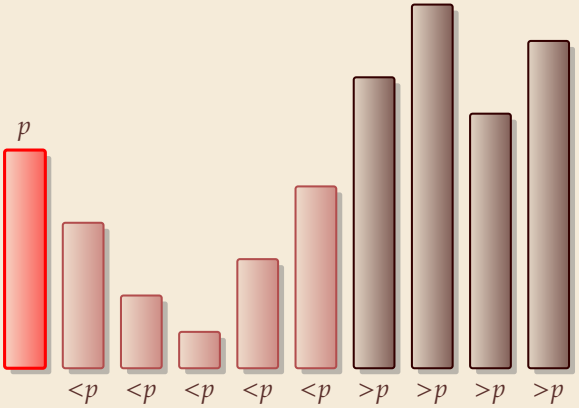
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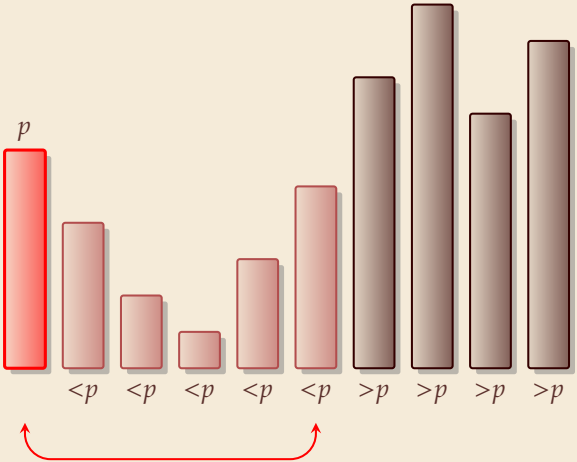
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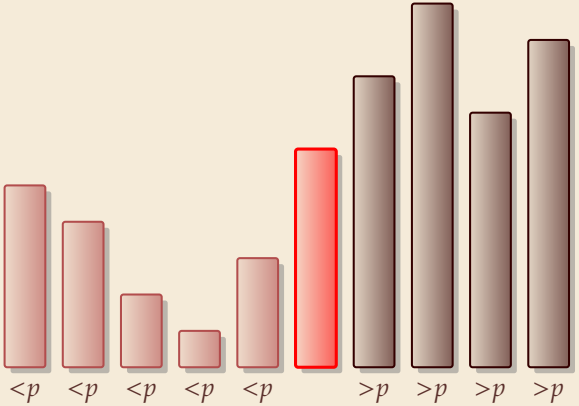
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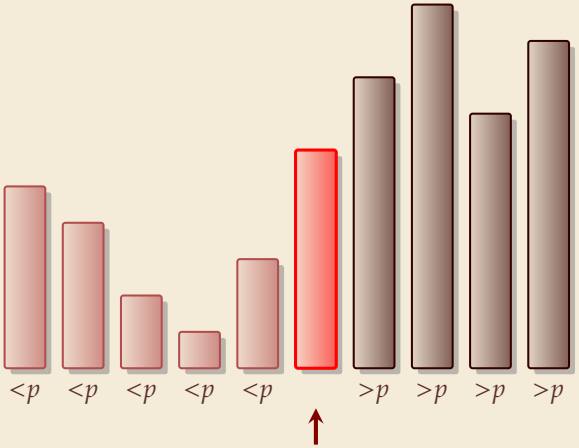
Partitioning around a pivot





Partitioning around a pivot





- no extra space needed
- visits each element once
- returns rank/position of pivot

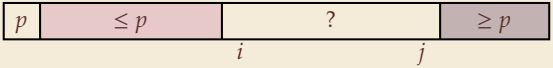
Partitioning – Detailed code

Beware: details easy to get wrong; use this code!

(if you ever have to)

```
1  procedure partition(A, b):
2      // input: array A[0..n), position of pivot b ∈ [0..n)
3      swap(A[0], A[b])
4      i := 0,    j := n
5      while true do
6          do i := i + 1 while i < n and A[i] < A[0]
7          do j := j − 1 while j ≥ 1 and A[j] > A[0]
8          if i ≥ j then break     (goto 11)
9          else swap(A[i], A[j])
10     end while
11     swap(A[0], A[j])
12     return j
```

Loop invariant (5–10): A



Quicksort

```
1 procedure quicksort(A[l..r)):
2     if r − l ≤ 1 then return
3     b := choosePivot(A[l..r))
4     j := l + partition(A[l..r), b)
5     quicksort(A[l..j))
6     quicksort(A[j + 1..r))
```

- recursive procedure
- choice of pivot can be
  - fixed position ⇝ dangerous!
  - random
  - more sophisticated, e. g., median of 3

Clicker Question

What is the worst-case running time of quicksort?



- **A** $\Theta(1)$
- **B** $\Theta(\log n)$
- **C** $\Theta(\log \log n)$
- **D** $\Theta(\sqrt{n})$
- **E** $\Theta(n)$
- **F** $\Theta(n \log \log n)$
- **G** $\Theta(n \log n)$
- **H** $\Theta(n \log^2 n)$
- **I** $\Theta(n^{1+\epsilon})$
- **J** $\Theta(n^2)$
- **K** $\Theta(n^3)$
- **L** $\Theta(2^n)$



→ *sli.do/effalg*

Clicker Question

What is the worst-case running time of quicksort?



A	~~$\Theta(1)$~~	G	~~$\Theta(n \log n)$~~
B	~~$\Theta(\log n)$~~	H	~~$\Theta(n \log^2 n)$~~
C	~~$\Theta(\log \log n)$~~	I	~~$\Theta(n^{1+\epsilon})$~~
D	~~$\Theta(\sqrt{n})$~~	J	$\Theta(n^2)$ ✓
E	~~$\Theta(n)$~~	K	~~$\Theta(n^3)$~~
F	~~$\Theta(n \log \log n)$~~	L	~~$\Theta(2^n)$~~



→ *sli.do/effalg*

Quicksort & Binary Search Trees

Quicksort



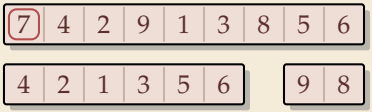
Quicksort & Binary Search Trees

Quicksort



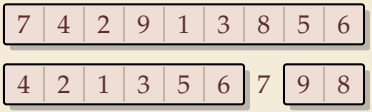
Quicksort & Binary Search Trees

Quicksort



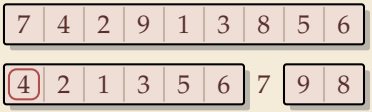
Quicksort & Binary Search Trees

Quicksort



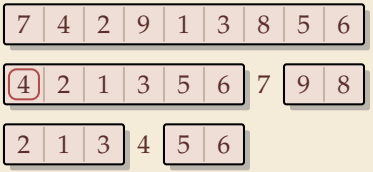
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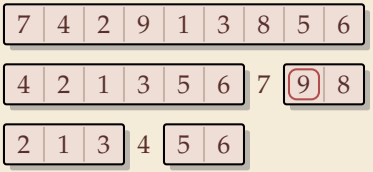
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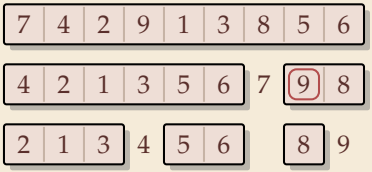
Quicksort & Binary Search Trees

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Quicksort & Binary Search Trees

Quicksort

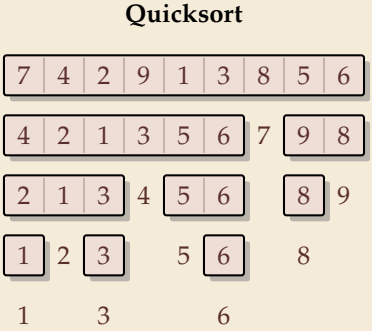


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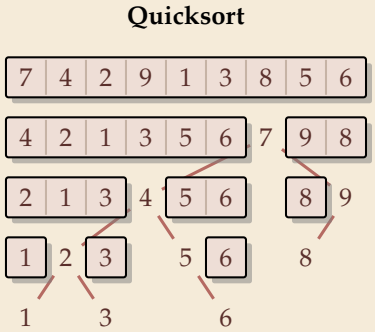
Quicksort



Quicksort & Binary Search Trees

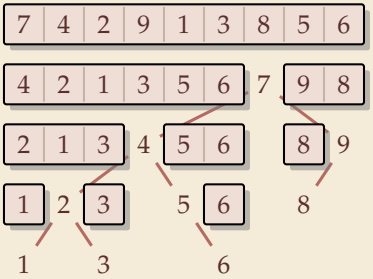


Quicksort & Binary Search Trees



Quicksort & Binary Search Trees

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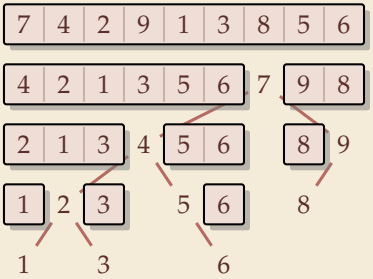


Binary Search Tree (BST)

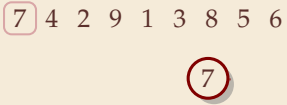
7 4 2 9 1 3 8 5 6

Quicksort & Binary Search Trees

Quicksort

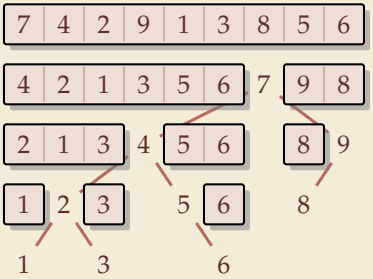


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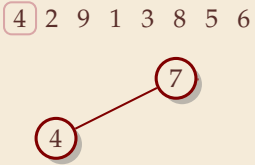


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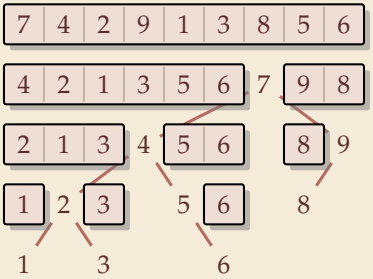


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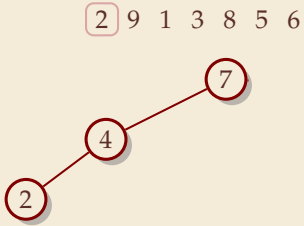


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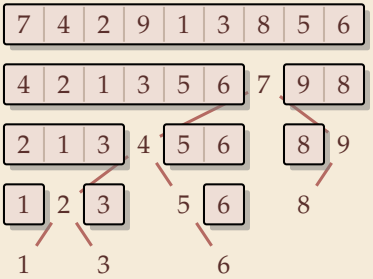


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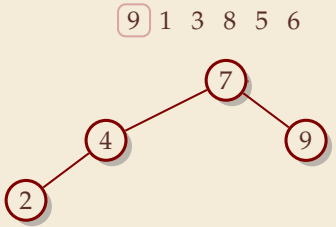


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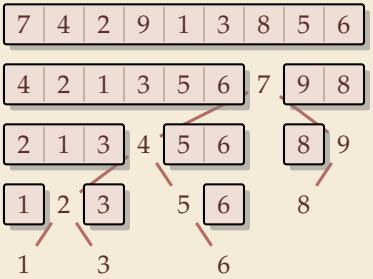


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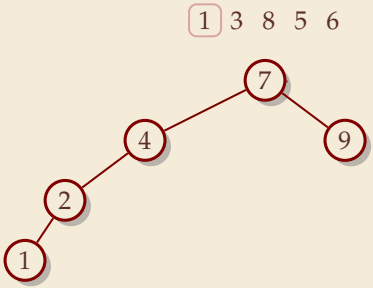


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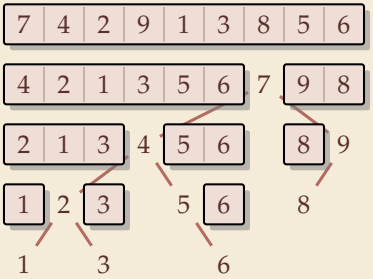


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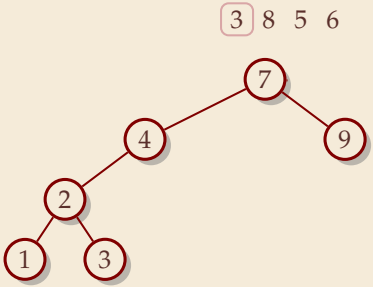


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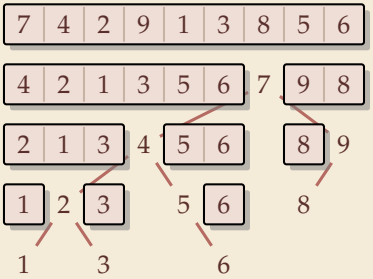


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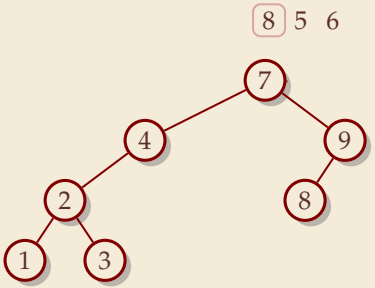


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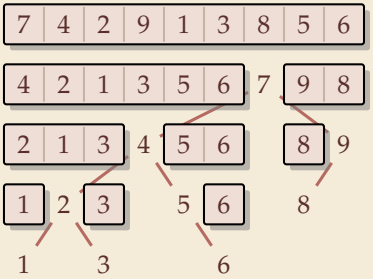


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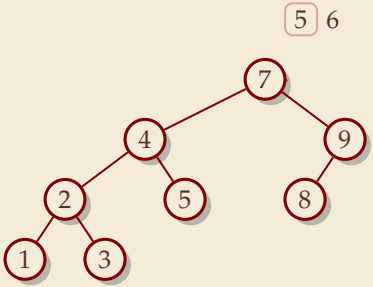


Quicksort & Binary Search Trees

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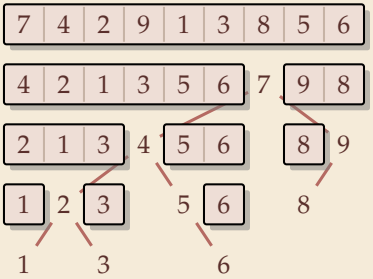


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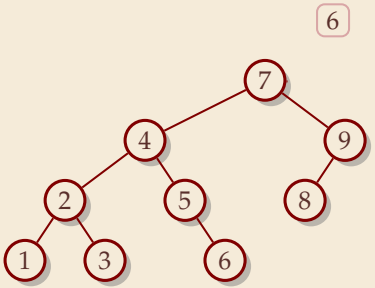


Quicksort & Binary Search Trees

Quicksort



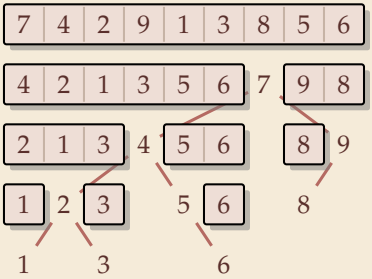
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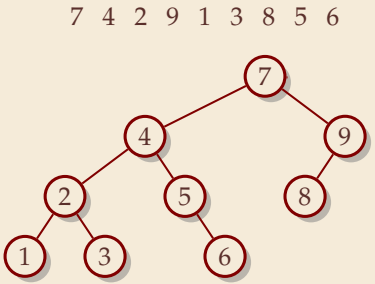
Quicksort & Binary Search Trees

(stable partitioning method)

Quicksort



Binary Search Tree (BST)

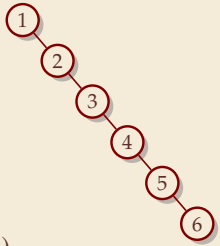


- recursion tree of quicksort = binary search tree from successive insertion
- comparisons in quicksort = comparisons to built BST
- comparisons in quicksort $\approx$ comparisons to search each element in BST

Quicksort – Worst Case

- Problem: BSTs can degenerate
- Cost to search for k is k

⇝ Total cost $\sum_{k=1}^{n} k = \frac{n(n+1)}{2} \sim \frac{1}{2}n^2$



⇝ quicksort worst-case running time is in $\Theta(n^2)$

terribly slow!

But, we can fix this:

Randomized quicksort:

- choose a *random pivot* in each step

⇝ same as randomly *shuffling* input before sorting

Randomized Quicksort – Analysis

- cost measure: element visits (as for mergesort)
- $C(n)$ = #element visits when sorting n randomly permuted elements
  = cost of searching every element in BST built from input

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$\rightsquigarrow$ quicksort needs $\sim 2\ln(2) \cdot n \lg n \approx 1.39 n \lg n$ *in expectation*
(see analysis of C_n in Unit 3!)

- also: very unlikely to be much worse:
  e. g., one can prove: $\Pr[\text{cost} > 10n \lg n] = O(n^{-2.5})$
  distribution of costs is "concentrated around mean"
  - even stronger bounds known  **McDiarmid, Hayward**: *Large Deviations for Quicksort*, J. Algorithms 21(3) **1996**
- intuition: have to be *constantly* unlucky with pivot choice

Quicksort – Discussion

- fastest general-purpose method
- $\Theta(n \log n)$ average case
- works *in-place* (no extra space required)
- memory access is sequential (scans over arrays)
- $\Theta(n^2)$ worst case (although extremely unlikely)
- not a *stable* sorting method

stable = elements equal with sorting order keep same relative ordering as in input

Open problem: Simple algorithm that is fast, stable and in-place.

4.3 Comparison-Based Lower Bound

Lower Bounds

- **Lower bound:** mathematical proof that *no algorithm* can do better.
  - very powerful concept: bulletproof *impossibility* result
    ≈ *conservation of energy* in physics
  - **(unique?) feature of computer science:**
    for many problems, solutions are known that (asymptotically) **achieve the lower bound**

  ⇝ can speak of "*optimal* algorithms"

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  ⇝ can speak of "*optimal* algorithms"

- To prove a statement about *all algorithms*, we must precisely define what that is!
- already know one option: the word-RAM model
- Here: use a simpler, more restricted model.

The Comparison Model

- In the *comparison model* data can only be accessed in two ways:
  - comparing two elements
  - moving elements around (e. g. copying, swapping)
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Keeps algorithms general!

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- Mergesort and Quicksort work in the comparison model.

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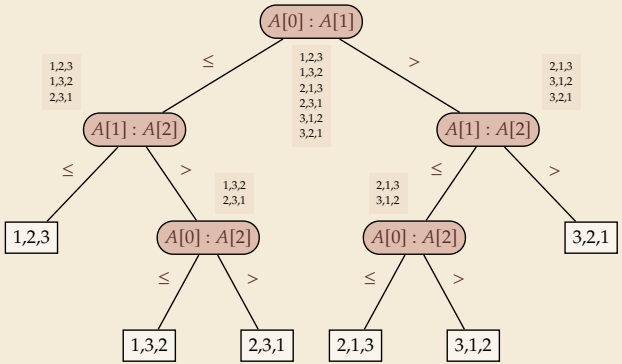
- This makes very few assumptions on the kind of objects we are sorting.
- Mergesort and Quicksort work in the comparison model.

⇝ Every comparison-based sorting algorithm corresponds to a ***decision tree***.

- only model comparisons ⇝ ignore data movement
- nodes = comparisons the algorithm does
- child links = outcomes of comparison
- leaf = unique initial input permutation compatible with comparison outcomes
- next comparisons can depend on outcomes ⇝ child subtrees can look different

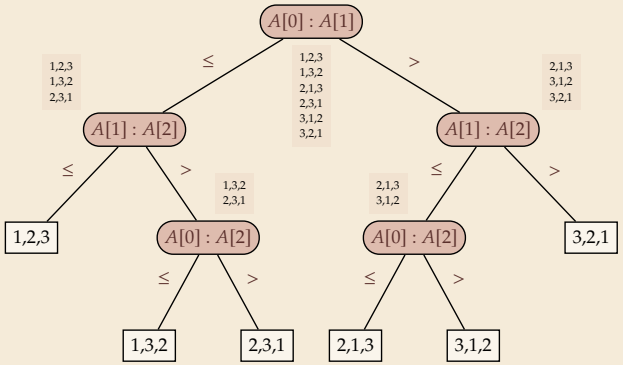
Comparison Lower Bound

Example: Comparison tree for a sorting method for $A[0..2]$:



Comparison Lower Bound

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- Execution = follow a path in comparison tree.

⇝ height of comparison tree = worst-case # comparisons

- comparison trees are *binary* trees

⇝ ℓ leaves ⇝ height $\geq \lceil \lg(\ell) \rceil$

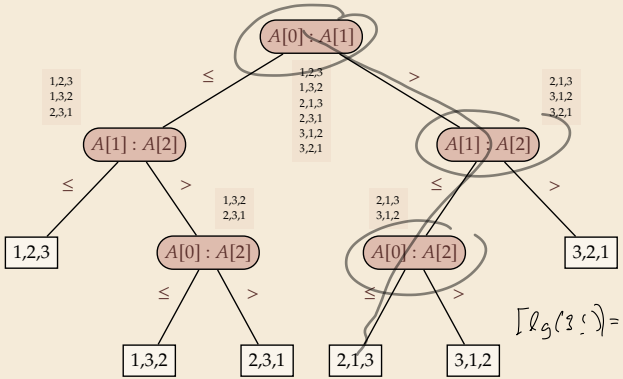
- comparison trees for sorting method must have $\geq n!$ leaves

⇝ height $\geq \lg(n!) \sim n \lg n$

more precisely: $\lg(n!) = n \lg n - \lg(e)n + O(\log n)$

Comparison Lower Bound

Example: Comparison tree for a sorting method for $A[0..2]$:



$\lceil \lg(3!) \rceil = \lceil \lg(6) \rceil = 3$

- Execution = follow a path in comparison tree.
  - ⇝ height of comparison tree = worst-case # comparisons
- comparison trees are *binary* trees
  - ⇝ ℓ leaves ⇝ height $\geq \lceil \lg(\ell) \rceil$
- comparison trees for sorting method must have $\geq n!$ leaves
  - ⇝ height $\geq \lg(n!) \sim n \lg n$

more precisely: $\lg(n!) = n \lg n - \lg(e)n + O(\log n)$

- Mergesort achieves $\sim n \lg n$ comparisons ⇝ asymptotically comparison-optimal!

Clicker Question



Does the comparison-tree from the previous slide correspond to a worst-case optimal sorting method?

A Yes

B No



→ *sli.do/effalg*

Clicker Question



Does the comparison-tree from the previous slide correspond to a worst-case optimal sorting method?

A Yes ✓

B ~~No~~



→ *sli.do/effalg*

How tight is the lower bound?

∉ exam

Worst-case #comparisons to sort n elements:

	comparisons	$\approx n \lg n - c \cdot n$
Lower bound $\lceil \lg n! \rceil$	$n \lg n - n \lg e + O(\log n)$	$c = \lg e \approx 1.443$
Mergesort (top-down)	$n\lceil \lg n \rceil - 2^{\lceil \lg n \rceil} + 1$	$c \in [0.91, 1]$
Binary insertion (#cmps only!)	$\sum_{k=1}^{n} \lceil \lg k \rceil = n\lceil \lg n \rceil - 2^{\lceil \lg n \rceil} + 1$	the same!
Merge-insertion (Ford–Johnson)	$n \lg n - (3 - \lg 3)\, n + O(\log n)$ (for best n)	$c \in [1.329, 1.415]$
Group-insertion (Sergeev 2020/21)	$\lg n! + o(n)$	$c \to \lg e$

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Group-insertion (Sergeev 2020/21)	$\lg n! + o(n)$	$c \longrightarrow \lg e$

- Merge-insertion is *exactly* optimal for $n \leq 15$ and $n = 20, 21$, but *not* for $n = 47$ (Manacher 1979).
- In 2020, Sergeev announced an algorithm with $\lg n! + o(n)$ comparisons
  long and intricate proof, only asymptotically better
- the $O(n)$ gap is irrelevant for running time – but it is the oldest question about sorting.

Ford, Johnson: *A Tournament Problem*, Amer. Math. Monthly 66(5) **1959**

Manacher: *The Ford–Johnson Sorting Algorithm Is Not Optimal*, J. ACM 26(3) **1979**

Sergeev: *On the Upper Bound of the Complexity of Sorting*, Comput. Math. Math. Phys. 61(2) / ECCC TR20-096 **2021**

4.4 Integer Sorting

Clicker Question

Select all **correct formulations** of our **lower bound** from §4.3.



A Any sorting algorithm requires $O(n \log n)$ running time in the worst case.

B Every comparison-based sorting algorithm requires $\Omega(n \log n)$ running time in the worst case for sorting n elements.

C Every comparison-based sorting algorithm requires $\Omega(n \log n)$ comparisons in the worst case for sorting n elements.

D Every sorting algorithm requires $\Omega(n \log n)$ comparisons in the worst case for sorting n elements.

E The complexity of sorting n elements in the comparison-model is $\Theta(n \log n)$.

F The complexity of sorting n elements in the comparison-model is $\Omega(n \log n)$.



→ *sli.do/effalg*

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D ~~Every sorting algorithm requires $\Omega(n \log n)$ comparisons in the worst case for sorting n elements.~~

E ~~The complexity of sorting n elements in the comparison-model is $\Theta(n \log n)$.~~

F The complexity of sorting n elements in the comparison-model is $\Omega(n \log n)$. ✓



→ *sli.do/effalg*

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- Here: sort n **integers**
  - can do *a lot* with integers: add them up, compute averages, ... (full power of word-RAM)
  - ⇝ we are **not** working in the comparison model
  - ⇝ *above lower bound does not apply!*

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  - but: a priori unclear how much arithmetic helps for sorting . . .

Counting sort

- Important parameter: size/range of numbers
  - numbers in range $[0..U) = \{0, \ldots, U-1\}$ typically $U = 2^b$ $\rightsquigarrow$ b-bit binary numbers

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  - numbers in range $[0..U) = \{0, \ldots, U-1\}$ typically $U = 2^b$ $\rightsquigarrow$ b-bit binary numbers
- We can sort n integers in $\Theta(n+U)$ time and $\Theta(U)$ space when $b \leq w$:

word size

Counting sort

```
1  procedure countingSort(A[0..n)):
2      // A contains integers in range [0..U).
3      C[0..U) := new integer array, initialized to 0
4      // Count occurrences
5      for i := 0,...,n-1
6          C[A[i]] := C[A[i]] + 1
7      i := 0 // Produce sorted list
8      for k := 0,...U-1
9          for j := 1,...C[k]
10             A[i] := k; i := i + 1
```

Θ(n)

all inner loops combined Θ(n)

- *count* how often each *possible* value occurs
- produce sorted result directly from counts
- circumvents lower bound by using integers as array index / pointer offset

$\rightsquigarrow$ Can sort n integers in range $[0..U)$ with $U = O(n)$ in time and space $\Theta(n)$.

Larger Universes: Radix Sort

- ***MSD Radix Sort:***
  - split numbers into base-R "digits"
  - Use counting sort on most significant digit
    (with variant of counting sort that moves full number)
  - $\rightsquigarrow$ integers sorted with respect to first digit
  - recurse on sublist for each digit value, using next digit for counting sort

$\rightsquigarrow$ After $\lfloor \log_R(U) \rfloor + 1$ levels of counting sort, fully sorted!

- For $R = O(n)$ (hence $R \leq 2^w$), all counting sort calls on the same level cost total of $O(n)$ time
  (requires care: reuse C across the small subproblems instead of re-initializing it)

$\rightsquigarrow$ total time $O(n \log_R(U)) = O\left(n \frac{\log(U)}{\log(R)}\right)$

$\rightsquigarrow$ $O(n)$ time sorting possible for numbers in range $U = O(n^c)$ for constant c.

Integer Sorting – State of the art

∉ exam

Algorithm theory

- integer sorting on the w-bit word-RAM
- suppose $U = 2^w$, but w can be an arbitrary function of n
- how fast can we sort n such w-bit integers on a w-bit word-RAM?
  - for $w = O(\log n)$: linear time (*radix/counting sort*)
  - for $w = \Omega(\log^{2+\varepsilon} n)$: linear time (*signature sort*)
  - for w in between: can do $O(n\sqrt{\lg \lg n})$ (randomized, very complicated algorithm)
    don't know if that is best possible!

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* * *

. . . for the rest of this unit: back to the comparisons model!

Clicker Question



Which statements are correct? Select all that apply.

My computer has 64-bit words, so an `int` has 64 bits. Hence I can sort any `int[]` of length n ...

- **A** in constant time.
- **B** in $O(\log n)$ time.
- **C** in $O(n)$ time.
- **D** in $O(n \log n)$ time.
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→ *sli.do/effalg*

Part II

Exploiting presortedness

4.5 Adaptive Sorting

Adaptive sorting

- Comparison lower bound also holds for the *average case* $\rightsquigarrow$ $\lg(n!)$ cmps necessary
- Mergesort and Quicksort from above use $\Theta(n \lg n)$ cmps even in best case

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Can we do better if the input is already "almost sorted"?

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Can we do better if the input is already "almost sorted"?

Scenarios where this may arise naturally:

- Append new data as it arrives, regularly sort entire list (e. g., log files, database tables)
- Compute summary statistics of time series of measurements that change slowly over time (e. g., weather data)
- Merging locally sorted data from different servers (e. g., map-reduce frameworks)

$\rightsquigarrow$ Ideally, algorithms should *adapt* to input: *the more sorted the input, the faster the algorithm*

. . . but how to do that!?

Warmup: check for sorted inputs

- Any method could first check if input already completely in order!
  - 👍 Best case becomes $\Theta(n)$ with $n - 1$ comparisons!
  - 👎 Usually $n - 1$ extra comparisons and pass over data "wasted"
  - 👎 Only catches a single, extremely special case ...

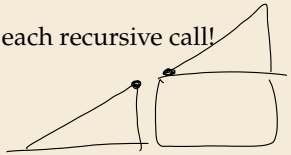
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For Mergesort, can instead check before merge with a **single** comparison

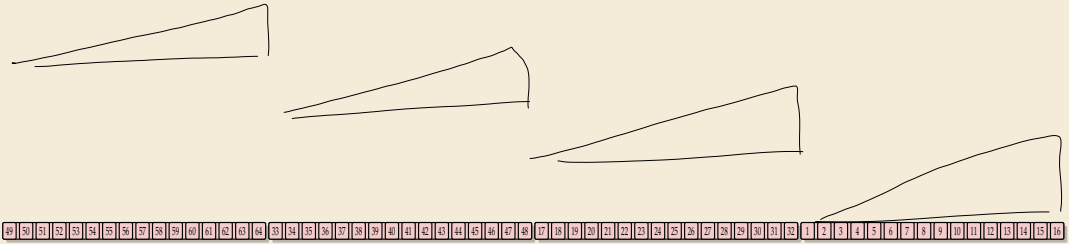
- If last element of first run $\leq$ first element of second run, skip merge

How effective is this idea?

```
1  procedure mergesortCheck(A[l..r)):
2      n := r − l
3      if n ≤ 1 return
4      m := l + ⌊n/2⌋
5      mergesortCheck(A[l..m))
6      mergesortCheck(A[m..r))
7      if A[m − 1] > A[m]
8          merge(A[l..m), A[m..r), buf)
9          copy buf to A[l..r)
```

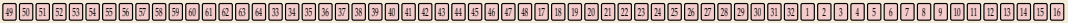
Mergesort with sorted check – Analysis

- Simplified cost measure: *merge cost* = size of output of merges
  ≈ number of comparisons
  ≈ number of memory transfers / cache misses
- Example input: $n = 64$ numbers in sorted *runs* of 16 numbers each:



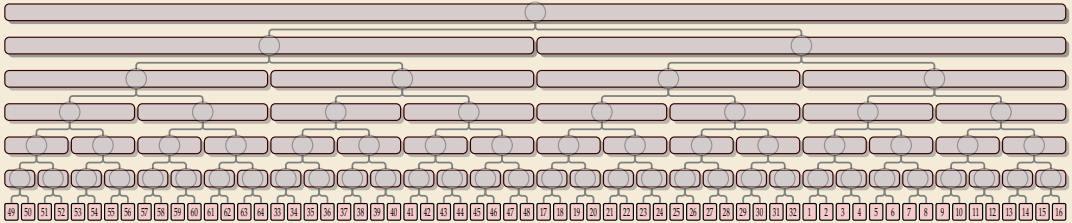
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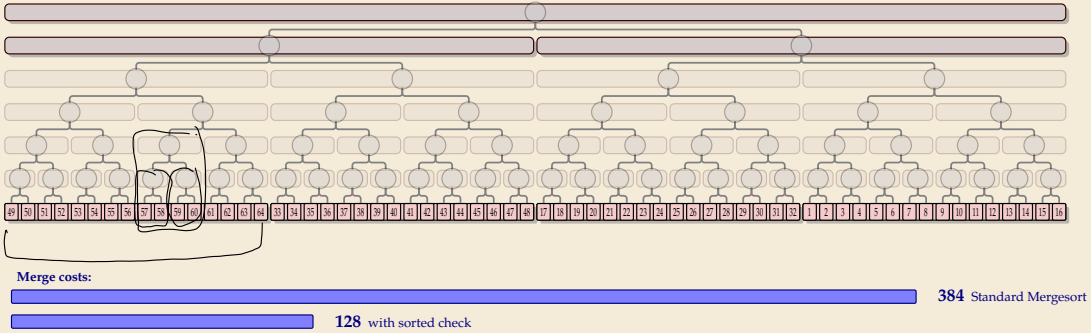
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384 Standard Mergesort

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Sorted check can help a lot!

Alignment issues

- In previous example, each run of length ℓ saved us $\ell \lg(\ell)$ in merge cost.
  - = exactly the cost of *creating* this run in mergesort had it not already existed
  - ⇝ best savings we can hope for!

⇝ Are overall merge costs $\mathcal{H}(\ell_1, \ldots, \ell_r) := \underbrace{n \lg(n)}_{\text{mergesort}} - \underbrace{\sum_{i=1}^{r} \ell_i \lg(\ell_i)}_{\text{savings from runs}}$?

(ℓ_i = length of ith run; $n = \sum \ell_i$)

$$= -\sum_{i=1}^{r} \left[\ell_i \lg(\ell_i) - \ell_i \lg n \right]$$

$$= \sum_{i=1}^{r} \ell_i \lg\left(\frac{n}{\ell_i}\right)$$

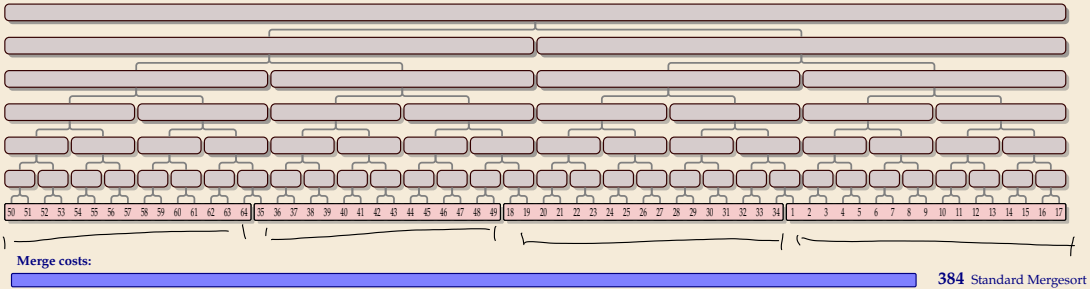
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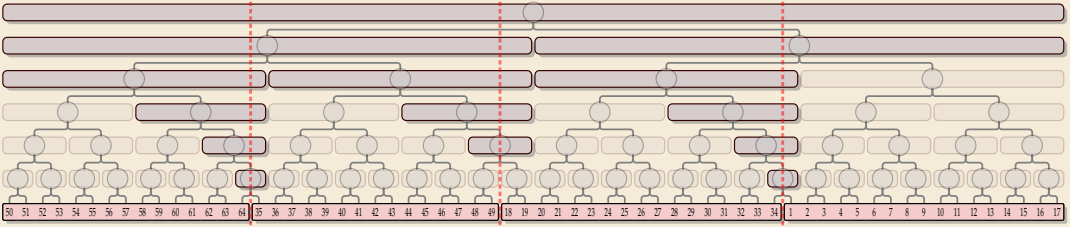
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Merge costs:

384 Standard Mergesort

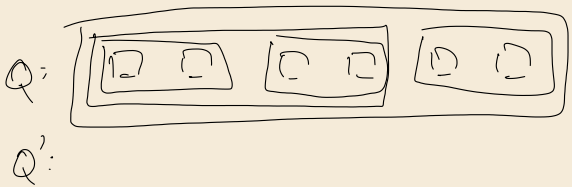
216 with sorted check

127.8 $\mathcal{H}(15, 15, 17, 17)$

Bottom-Up Mergesort

- Can we do better by explicitly detecting runs?

```
1  procedure bottomUpMergesort(A[0..n)):
2      Q := new Queue // runs to merge
3      // Phase 1: Enqueue singleton runs
4      for i = 0, ..., n − 1 do
5          Q.enqueue((i, i + 1))
6      // Phase 2: Merge runs level−wise
7      while Q.size() ≥ 2
8          Q′ := new Queue
9          while Q.size() ≥ 2
10             (i1, j1) := Q.dequeue()
11             (i2, j2) := Q.dequeue()
12             merge(A[i1..j1), A[i2..j2), buf)
13             copy buf to A[i1..j2)
14             Q′.enqueue((i1, j2))
15         if ¬Q.isEmpty() // lonely run
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```

```
1  procedure naturalMergesort(A[0..n)):
2      Q := new Queue;  i := 0
3      while i < n
4          j := i + 1
5          while j<n ∧ A[j] ≥A[j−1] do j:=j+1
6          Q.enqueue((i, j));  i := j
7      while Q.size() ≥ 2
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```

find run $A[i..j)$ starting at i

Clicker Question

Suppose we have an input with the 5 elements `a, b, c, d, e` and we sort them with **bottomUpMergesort**. What sequence of merges are executed?



A Policy 1

a	b	c	d	e
a	b	c	d	e
a	b	c	d	e
a	b	c	d	e

Policy 1

B Policy 2

a	b	c	d	e
a	b	c	d	e
a	b	c	d	e
a	b	c	d	e

Policy 2

C Policy 3

a	b	c	d	e
a	b	c	d	e
a	b	c	d	e
a	b	c	d	e

Policy 3



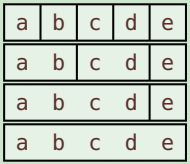
→ *sli.do/effalg*

Clicker Question

Suppose we have an input with the 5 elements `a, b, c, d, e` and we sort them with **bottomUpMergesort**. What sequence of merges are executed?

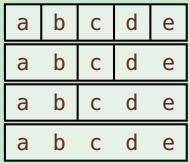


A Policy 1 ✓



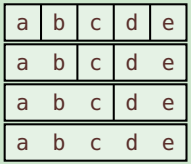
Policy 1

B ~~Policy 2~~



Policy 2

C ~~Policy 3~~



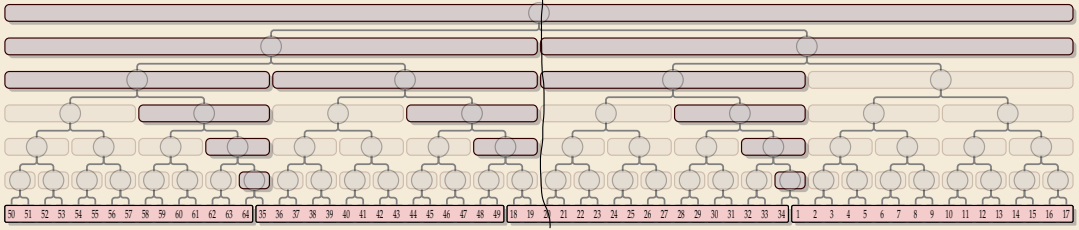
Policy 3



→ *sli.do/effalg*

Natural Bottom-Up Mergesort – Analysis

- Works well for runs of roughly equal size, regardless of alignment . . .

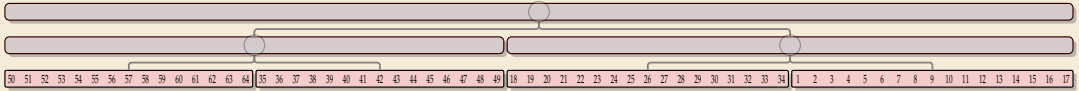


Merge costs:

384 Standard mergesort

216 Standard mergesort with sorted check

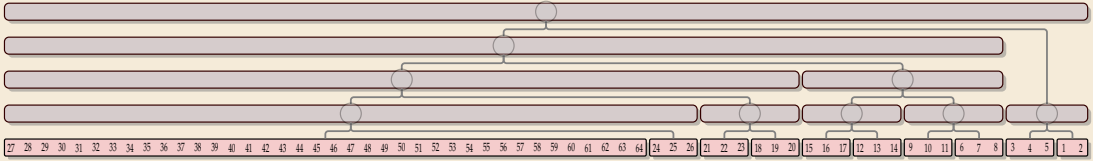
127.8 $\mathcal{H}(15, 15, 17, 17)$



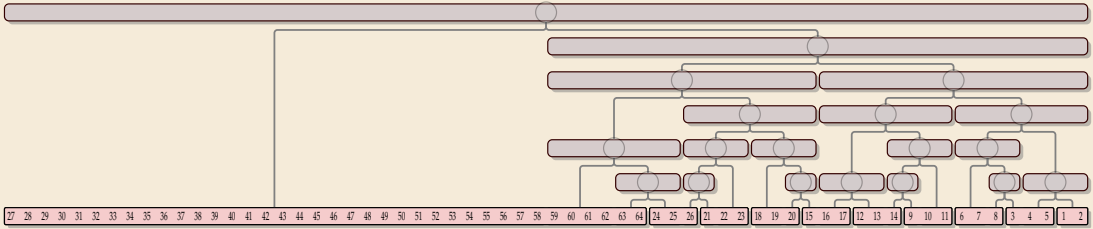
128 Natural bottom-up mergesort

Natural Bottom-Up Mergesort – Analysis [2]

- . . . but less so for widely varying run lengths



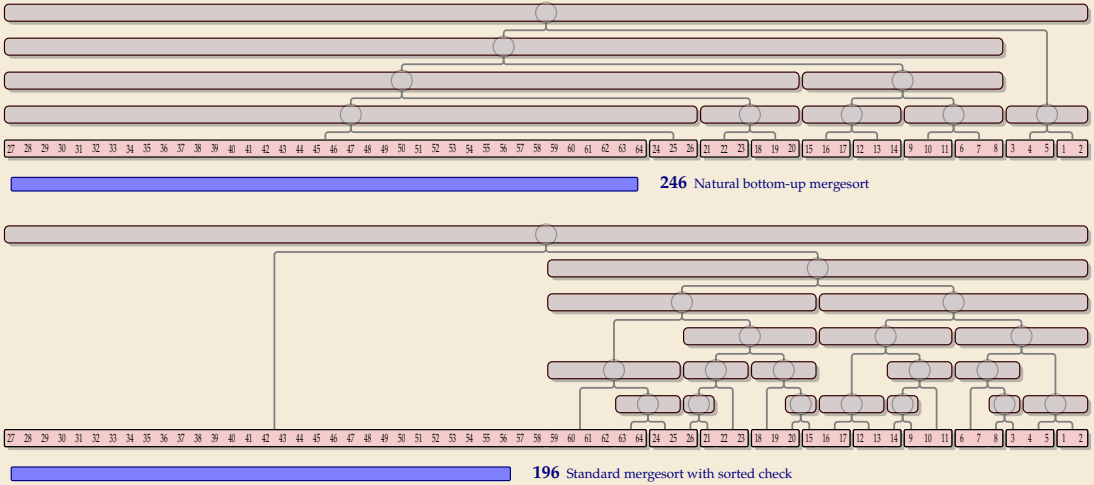
246 Natural bottom-up mergesort



196 Standard mergesort with sorted check

Natural Bottom-Up Mergesort – Analysis [2]

- . . . but less so for widely varying run lengths



. . . can't we have both at the same time?!

Good merge orders



Let's take a step back and breathe.

Good merge orders



Let's take a step back and breathe.

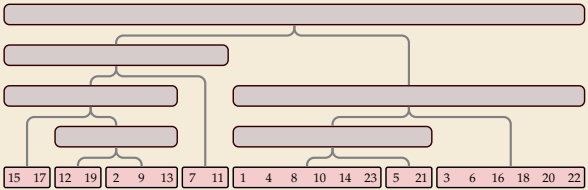
- Conceptually, there are two tasks:
  1. Detect and use existing runs in the input $\rightsquigarrow$ $\ell_1, \ldots, \ell_r$ (easy)
  2. Determine a favorable *order of merges* of runs ("automatic" in top-down mergesort)

Good merge orders



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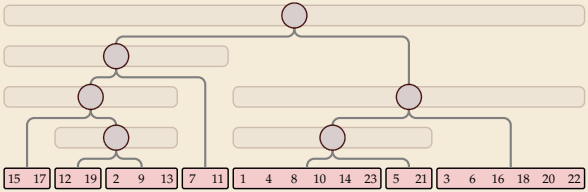


Merge cost = total area of 

Good merge orders

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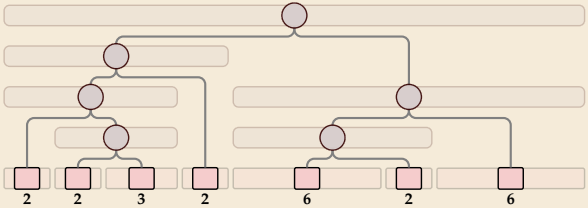


Merge cost = total area of 
= total length of paths to all array entries

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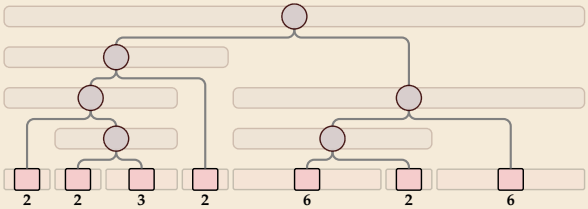
= total length of paths to all array entries

$$= \sum_{w \text{ leaf}} weight(w) \cdot depth(w)$$

Good merge orders

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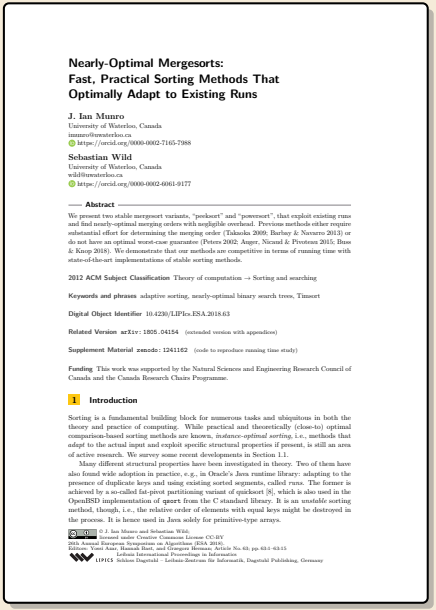
= total length of paths to all array entries

$$= \sum_{w \text{ leaf}} weight(w) \cdot depth(w)$$

$\rightsquigarrow$ *optimal* merge tree = optimal *binary search tree* for leaf weights $\ell_1, \ldots, \ell_r$ (optimal expected search cost)

well-understood problem with known algorithms

Nearly-Optimal Mergesort



Nearly-Optimal Mergesorts:
Fast, Practical Sorting Methods That Optimally Adapt to Existing Runs

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Abstract

We present two stable mergesort variants, "peeksort" and "powersort", that exploit existing runs and find nearly-optimal merging orders with negligible overhead. Previous methods either require substantial effort for determining the merging order (Takaoka 2009; Barbay & Navarro 2013) or do not have an optimal worst-case guarantee (Peters 2002; Auger, Nicaud & Pivoteau 2015; Buss & Knop 2018). We demonstrate that our methods are competitive in terms of running time with state-of-the-art implementations of stable sorting methods.

2012 ACM Subject Classification Theory of computation → Sorting and searching

Keywords and phrases adaptive sorting, nearly-optimal binary search trees, Timsort

Digital Object Identifier 10.4230/LIPIcs.ESA.2018.63

Related Version `arXiv:1805.04154` (extended version with appendices)

Supplement Material `zenodo:1241162` (code to reproduce running time study)

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1 Introduction

Sorting is a fundamental building block for numerous tasks and ubiquitous in both the theory and practice of computing. While practical and theoretically (close-to) optimal comparison-based sorting methods are known, *instance-optimal sorting*, i.e., methods that *adapt* to the actual input and exploit specific structural properties if present, is still an area of active research. We survey some recent developments in Section 1.1.

Many different structural properties have been investigated in theory. Two of them have also found wide adoption in practice, e.g., in Oracle's Java runtime library: adapting to the presence of duplicate keys and using existing sorted segments, called *runs*. The former is achieved by a so-called fat-pivot partitioning variant of quicksort [8], which is also used in the OpenBSD implementation of `qsort` from the C standard library. It is an *unstable* sorting method, though, i.e., the relative order of elements with equal keys might be destroyed in the process. It is hence used in Java solely for primitive-type arrays.

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26th Annual European Symposium on Algorithms (ESA 2018).
Editors: Yossi Azar, Hannah Bast, and Grzegorz Herman; Article No. 63; pp. 63:1–63:15
Leibniz International Proceedings in Informatics
LIPICS Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany

- In 2018, with Ian Munro, I combined research on nearly-optimal BSTs with mergesort

⇝ 2 new algorithms: *Peeksort* and *Powersort*

- both adapt provably optimal to existing runs even in worst case:
  mergecost $\leq \mathcal{H}(\ell_1, \dots, \ell_r) + 2n$
- both are lightweight extensions of existing methods with negligible overhead
- both fast in practice

Munro, Wild: *Nearly-Optimal Mergesorts: Fast, Practical Sorting Methods That Optimally Adapt to Existing Runs*, ESA **2018**

Peeksort

- based on top-down mergesort
- "peek" at middle of array
  & find closest run boundary



⇝ split there and recurse
(instead of at midpoint)

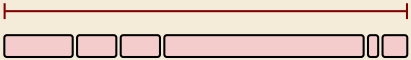


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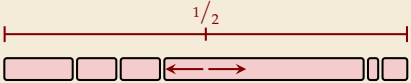


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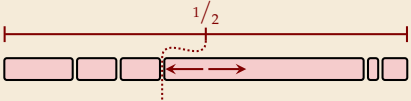
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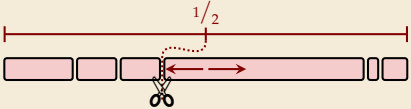
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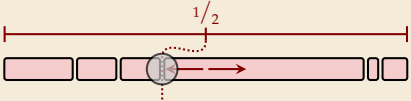
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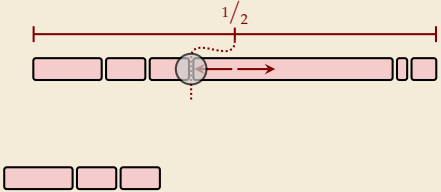


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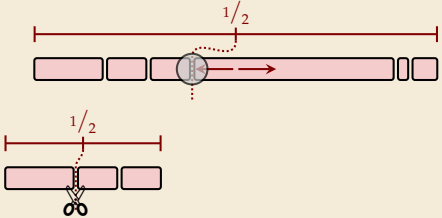


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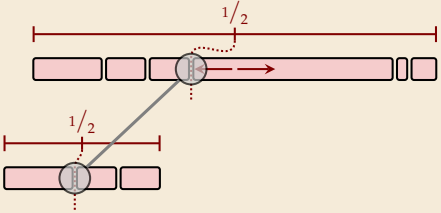


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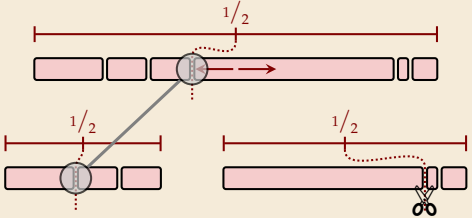


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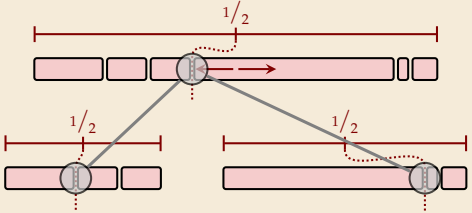


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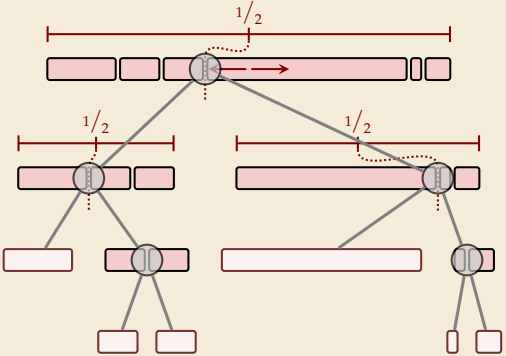


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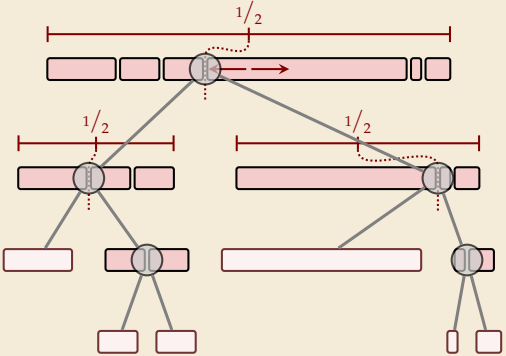


Peeksort

- based on top-down mergesort
- "peek" at middle of array & find closest run boundary



⇝ split there and recurse
(instead of at midpoint)



- can avoid scanning runs repeatedly:
  - find full run straddling midpoint
  - remember length of known runs at boundaries



⇝ with clever recursion, scan each run only once.

Peeksort – Code

1 **procedure** peeksort($A[\ell..r)$, Δ_ℓ, Δ_r):
2 **if** $r - \ell \leq 1$ **then return**
3 **if** $\ell + \Delta_\ell == r \vee \ell == r + \Delta_r$ **then return**
4 $m := \ell + \lfloor (r - \ell)/2 \rfloor$
5 $i := \begin{cases} \ell + \Delta_\ell & \text{if } \ell + \Delta_\ell \geq m \\ \text{extendRunLeft}(A, m) & \text{else} \end{cases}$
6 $j := \begin{cases} r + \Delta_r & \text{if } r + \Delta_r \leq m \\ \text{extendRunRight}(A, m) & \text{else} \end{cases}$
7 $g := \begin{cases} i & \text{if } m - i < j - m \\ j & \text{else} \end{cases}$
8 $\Delta_g := \begin{cases} j - i & \text{if } m - i < j - m \\ i - j & \text{else} \end{cases}$
9 peeksort($A[\ell..g)$, Δ_ℓ, Δ_g)
10 peeksort($A[g, r)$, Δ_g, Δ_r)
11 merge($A[\ell, g)$, $A[g..r)$, *buf*)
12 copy *buf* to $A[\ell..r)$

- Parameters:



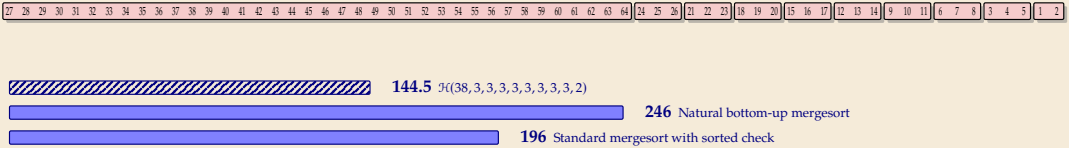
- initial call:
  peeksort($A[0..n)$, Δ_0, Δ_n) with
  $\Delta_0 = \text{extendRunRight}(A, 0)$
  $\Delta_n = \text{extendRunLeft}(A, n - 1) - n$
- helper procedure

1 **procedure** extendRunRight($A[0..n)$, i):
2 $j := i + 1$
3 **while** $j < n \wedge A[j - 1] \leq A[j]$
4 $j := j + 1$
5 **return** j // $A[i..j)$ *is a run*

(extendRunLeft similar)

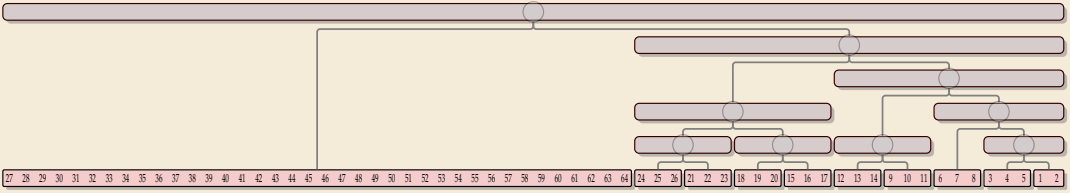
Peeksort – Analysis

- Consider tricky input from before again:



Peeksort – Analysis

- Consider tricky input from before again:



- One can prove: Mergecost always $\leq \mathcal{H}(\ell_1, \ldots, \ell_r) + 2n$

$\rightsquigarrow$ We can have the best of both worlds!

4.6 Python's list sort

Sorting in Python

- *CPython*
  - *Python* is only a *specification* of a programming language
  - The Python Foundation maintains *CPython* as the official reference implementation
  - If you don't specifically install something else, `python` will be CPython
- part of Python are `list.sort` resp. `sorted` built-in functions
  - implemented in C
  - use *Timsort*, custom Mergesort variant by Tim Peters

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Sept 2021: **Python uses *Powersort*!**

since CPython 3.11 and PyPy 7.3.6

Sorting in Python . . . and JavaScript

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April 2026: **Chrome's V8 too!**
`Array.prototype.sort`
in Chrome, Chromium, Edge, Node.js, . . .
since Chromium 149

Sorting in Python . . . and JavaScript

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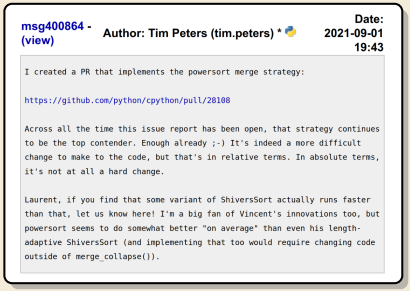


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April 2026: **Chrome's V8 too!**
`Array.prototype.sort`
in Chrome, Chromium, Edge, Node.js, . . .
since Chromium 149

also: Apple's WebKit (Safari), NumPy, AssemblyScript, . . .


msg400864 - (view) Author: Tim Peters (tim.peters) * Date: 2021-09-01 19:43

I created a PR that implements the powersort merge strategy:

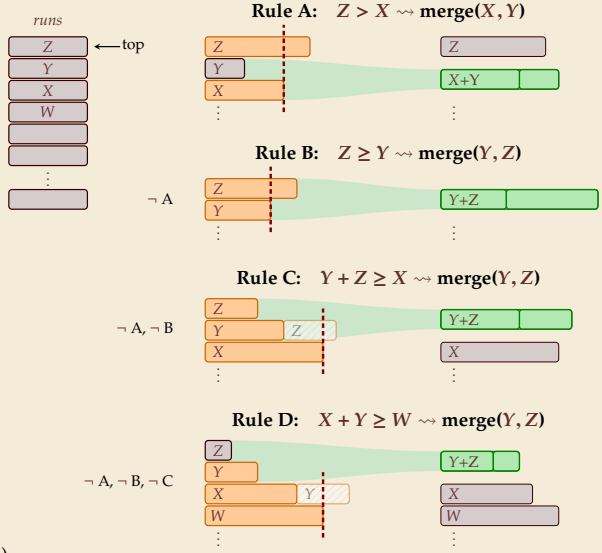
https://github.com/python/cpython/pull/28108

Across all the time this issue report has been open, that strategy continues to be the top contender. Enough already ;-) It's indeed a more difficult change to make to the code, but that's in relative terms. In absolute terms, it's not at all a hard change.

Laurent, if you find that some variant of ShiversSort actually runs faster than that, let us know here! I'm a big fan of Vincent's innovations too, but powersort seems to do somewhat better "on average" than even his length-adaptive ShiversSort (and implementing that too would require changing code outside of merge_collapse()).

Timsort (original version)

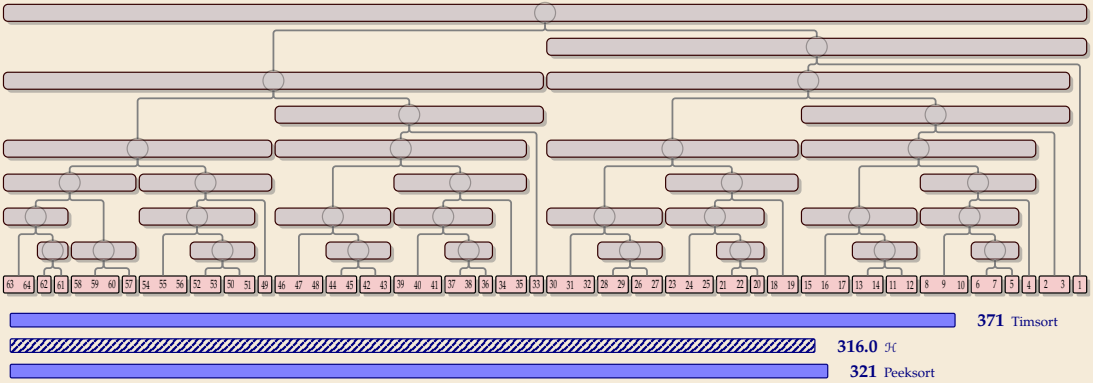
```
1 procedure Timsort(A[0..n)):
2     i := 0; runs := new Stack()
3     while i < n
4         j := ExtendRunRight(A, i)
5         runs.push(i, j); i := j
6         while rule A/B/C/D applicable
7             merge corresponding runs
8     while runs.size() ≥ 2
9         merge topmost 2 runs
```



- above shows the core algorithm;
  many more algorithm engineering tricks
- **Advantages:**
  - profits from existing runs
  - *locality of reference* for merges
- **But:** ***not*** optimally adaptive! (next slide)
  Reason: Rules A–D (Why exactly these?!)

Timsort bad case

- On certain inputs, Timsort's merge rules don't work well:

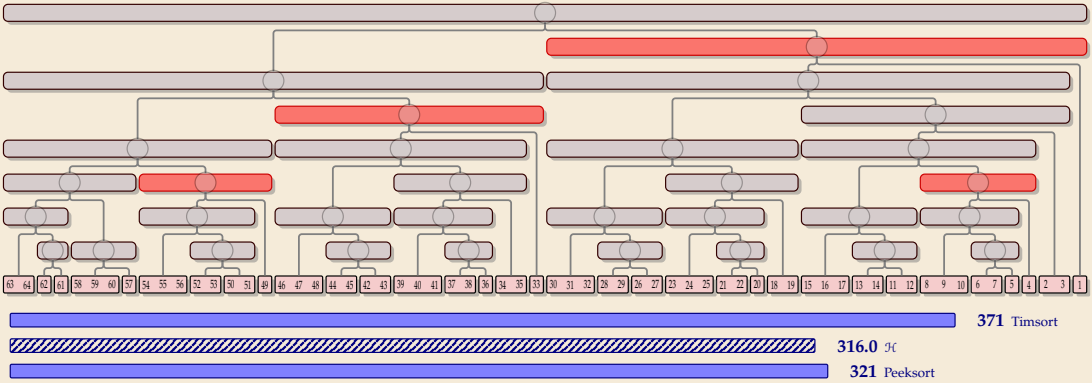


- As n increases, Timsort's cost approach $1.5 \cdot \mathcal{H}$, i. e., 50% more merge costs than necessary

Buss, Knop: *Strategies for Stable Merge Sorting*, SODA **2019**

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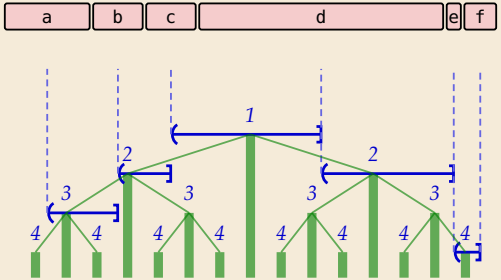
- As n increases, Timsort's cost approach $1.5 \cdot \mathcal{H}$, i. e., 50% more merge costs than necessary
  - intuitive problem: regularly very unbalanced merges

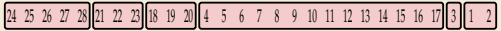
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Powersort

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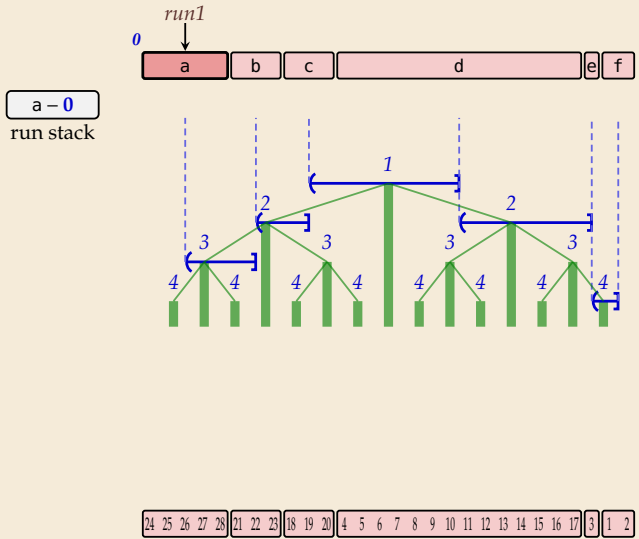




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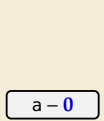
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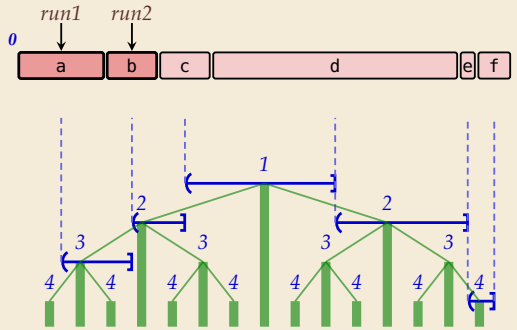


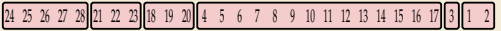
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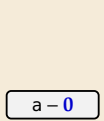


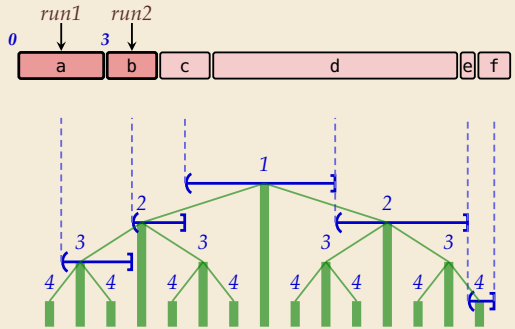


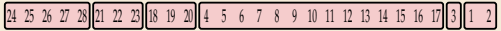
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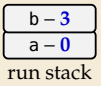


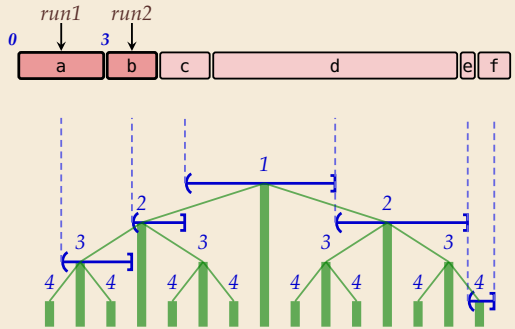


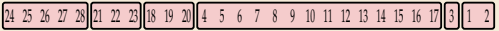
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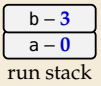


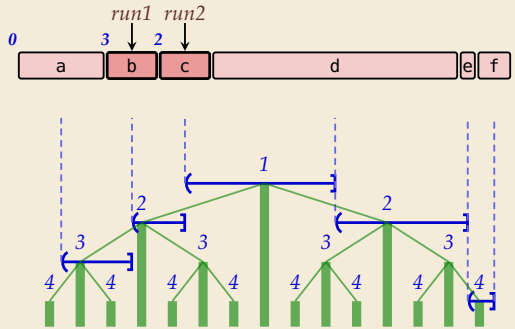


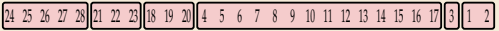
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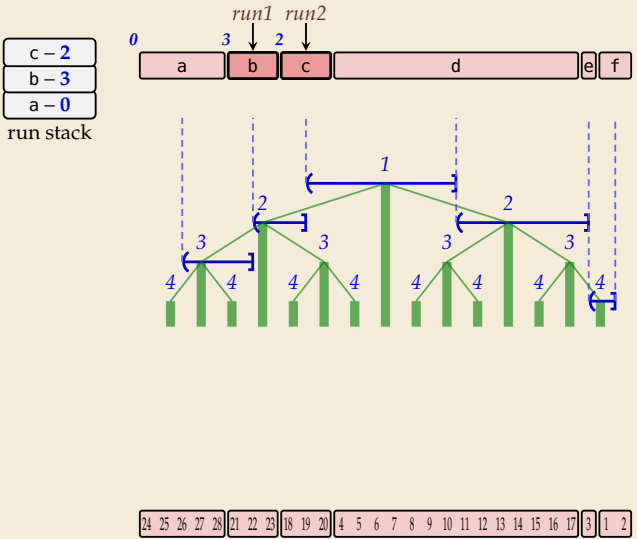




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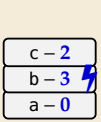
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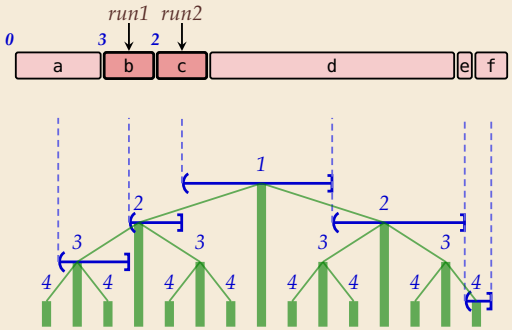


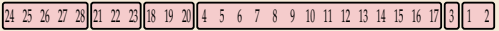
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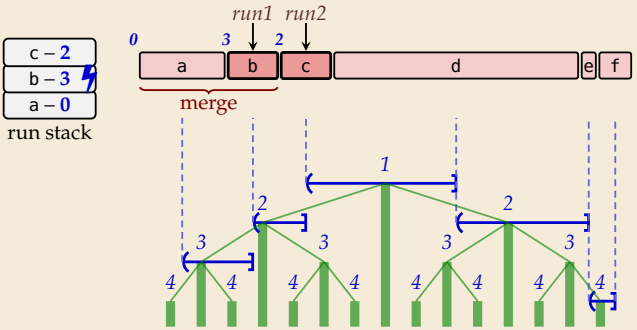


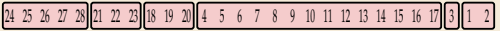


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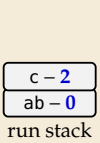


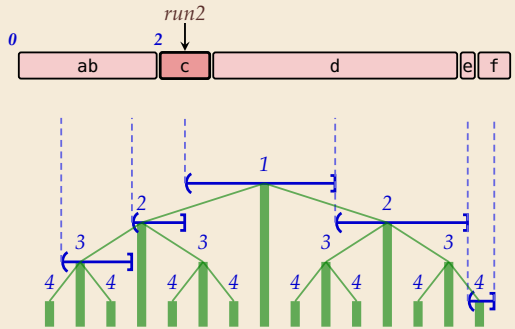


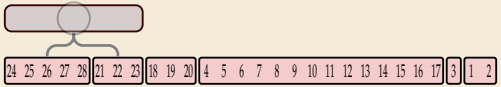
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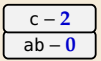


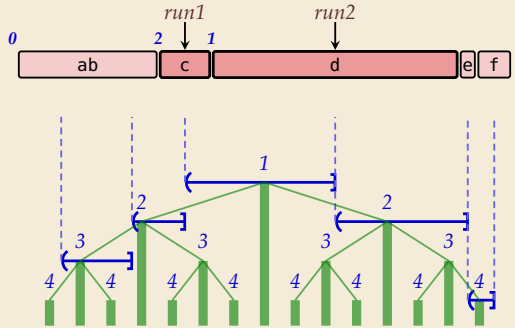


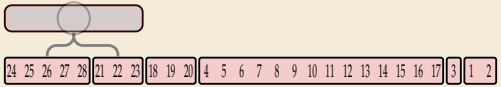
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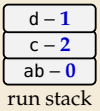


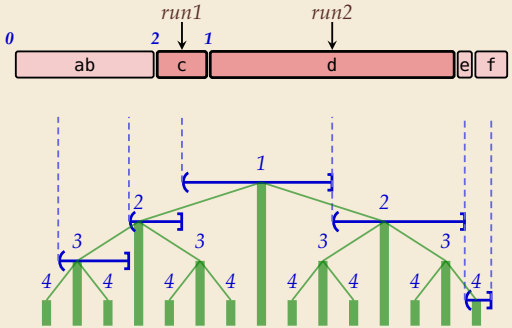


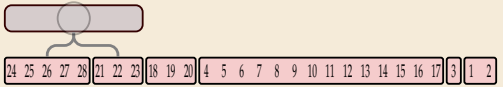
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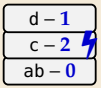


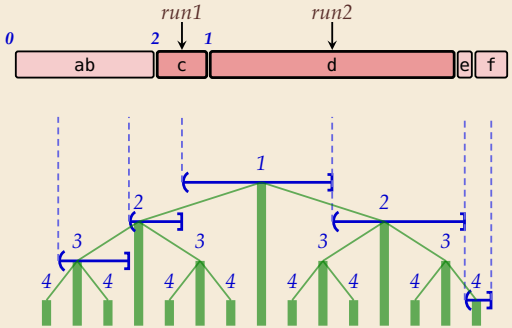


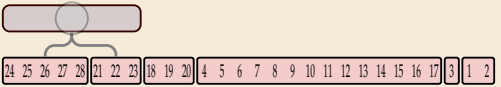
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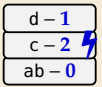


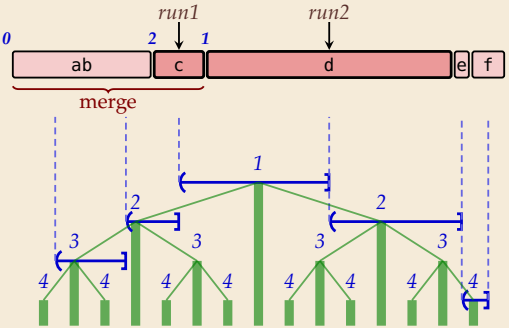


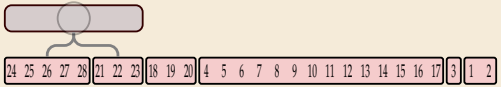
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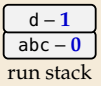


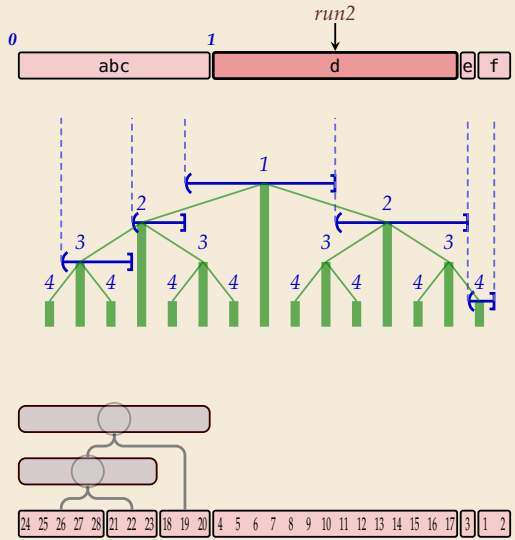


Powersort

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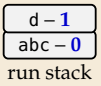


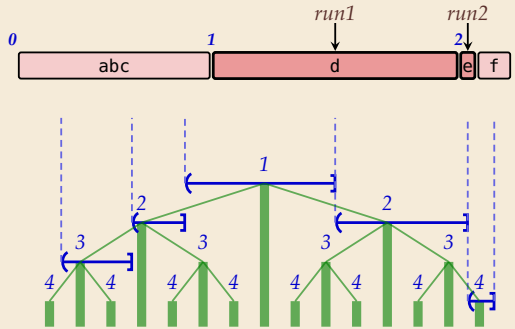


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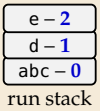


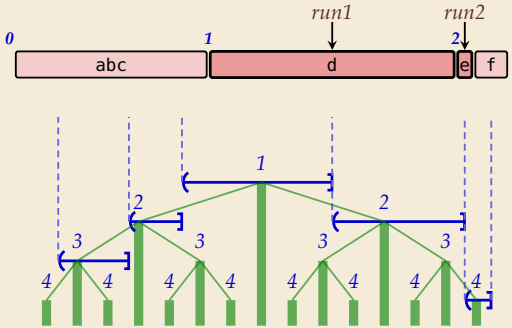


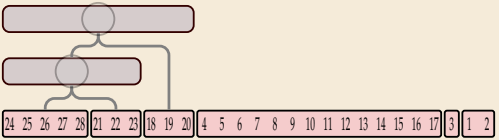
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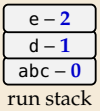


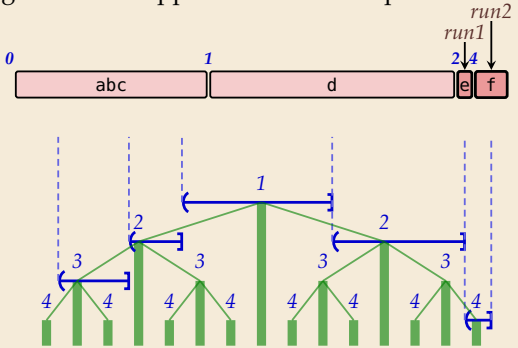


Powersort

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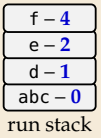


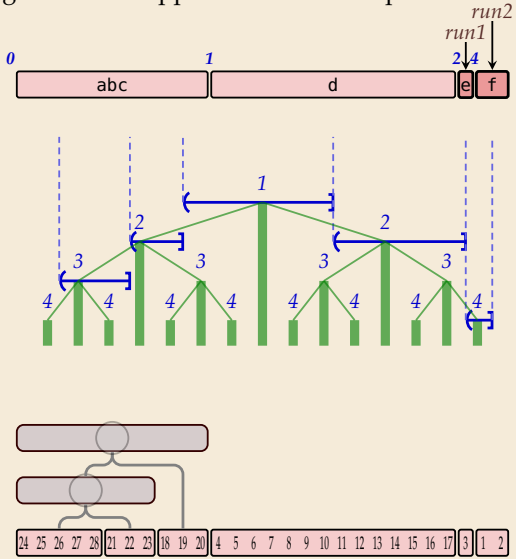


Powersort

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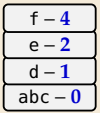




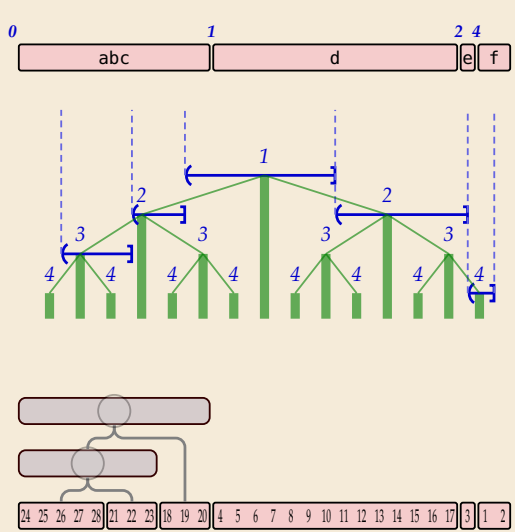
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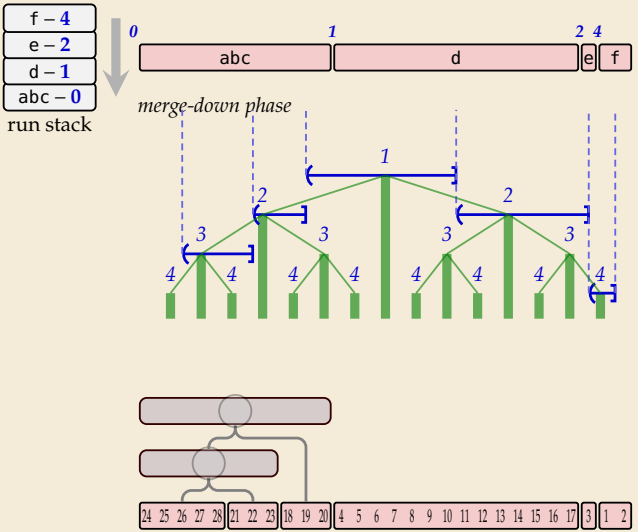
run stack



Powersort

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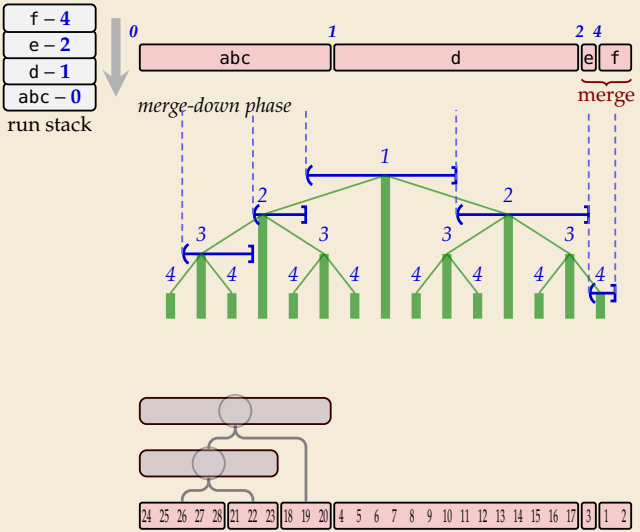
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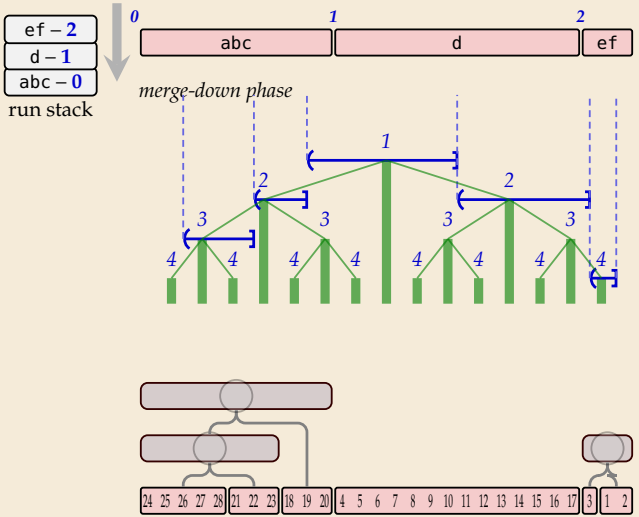
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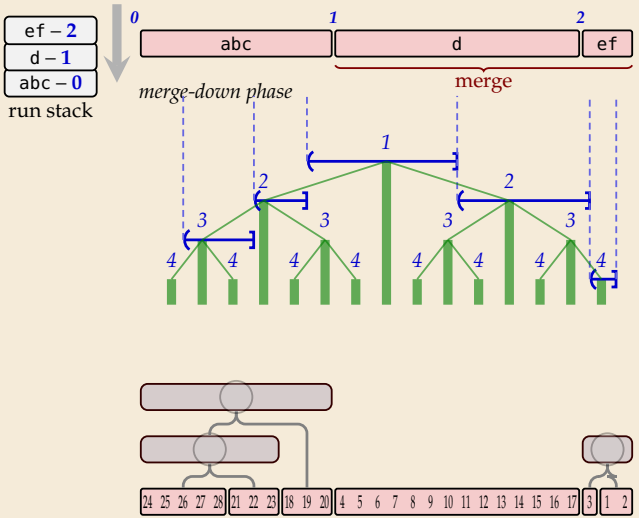
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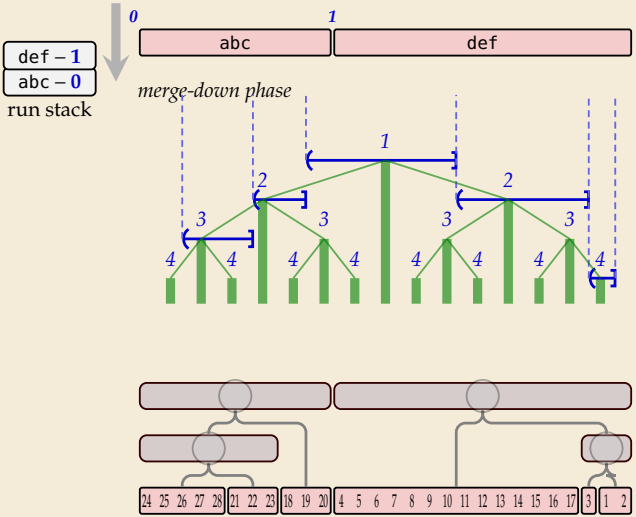
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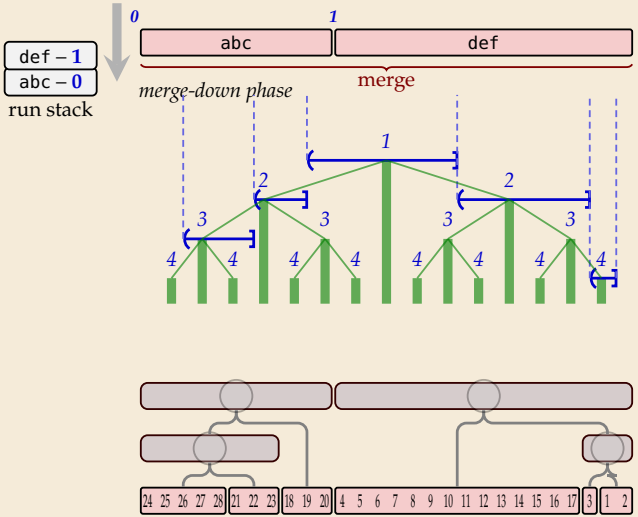
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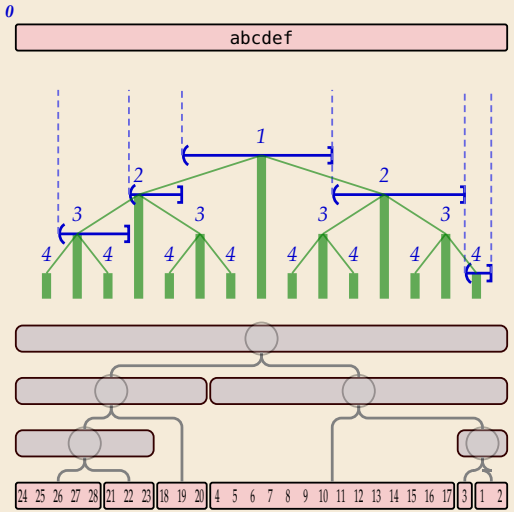


Powersort

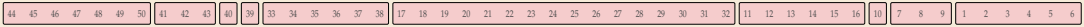
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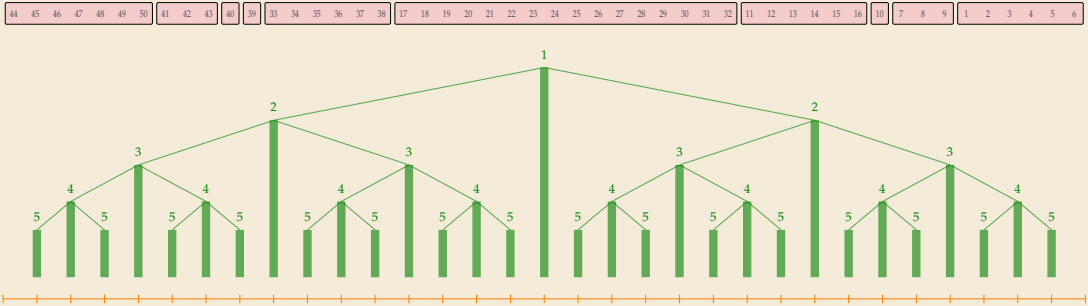
abcdef – 0
run stack



Powersort – Run-Boundary Powers

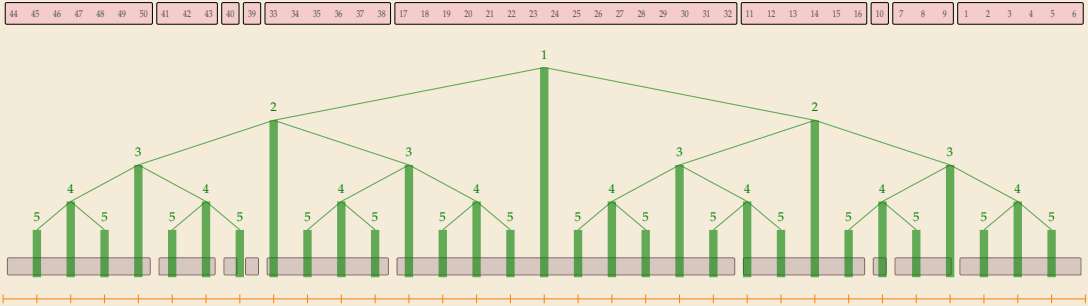


Powersort – Run-Boundary Powers



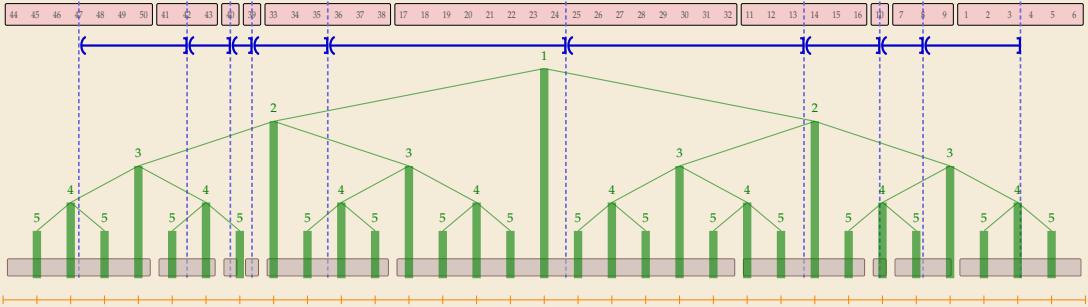
- (virtual) perfect balanced binary tree

Powersort – Run-Boundary Powers



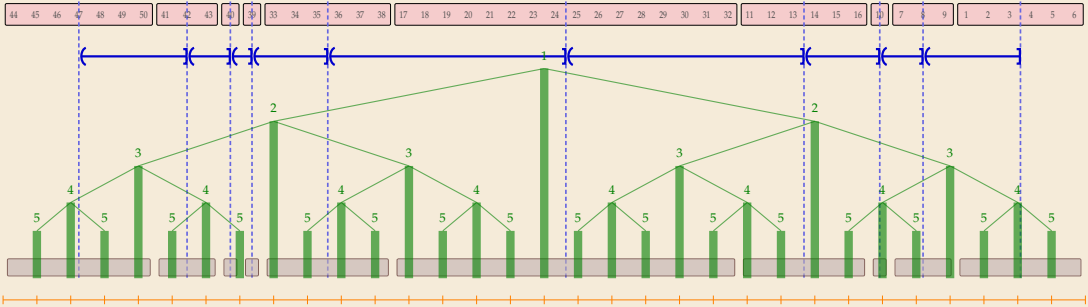
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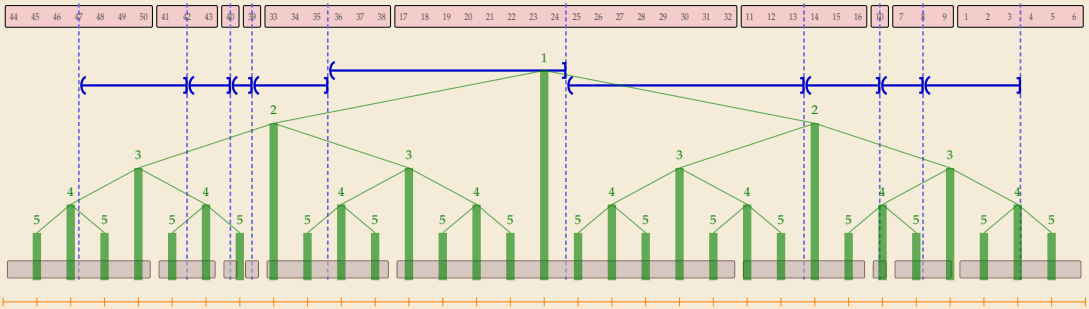
- (virtual) perfect balanced binary tree
- midpoint intervals “snap” to closest virtual tree node

Powersort – Run-Boundary Powers



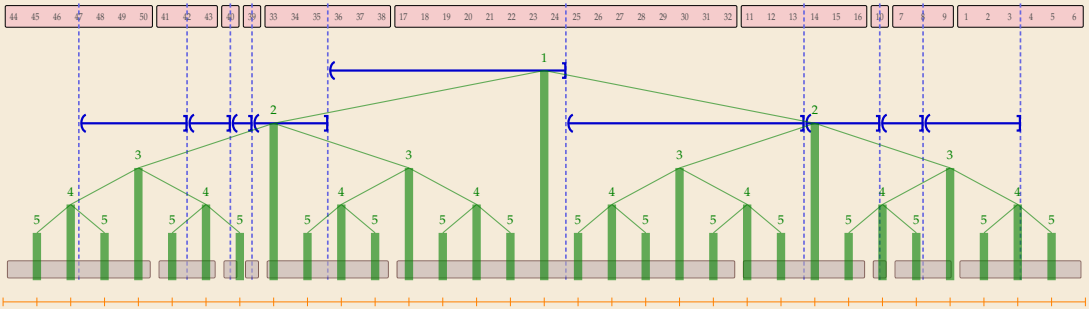
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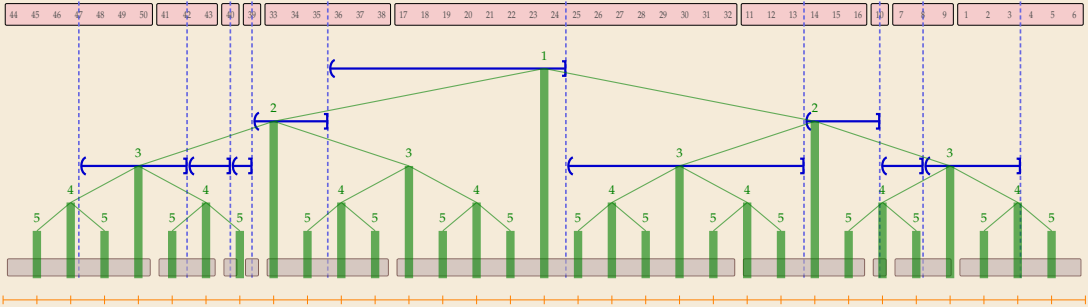
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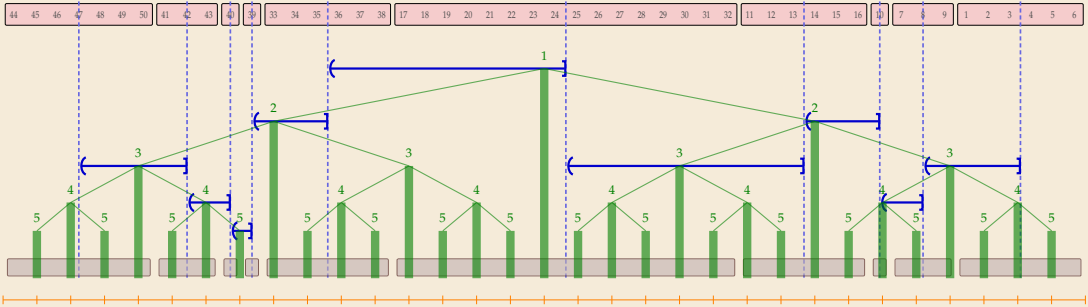
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Powersort – Run-Boundary Powers



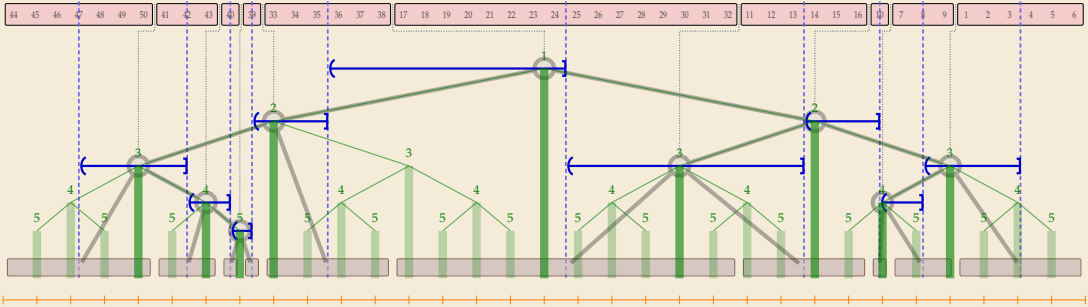
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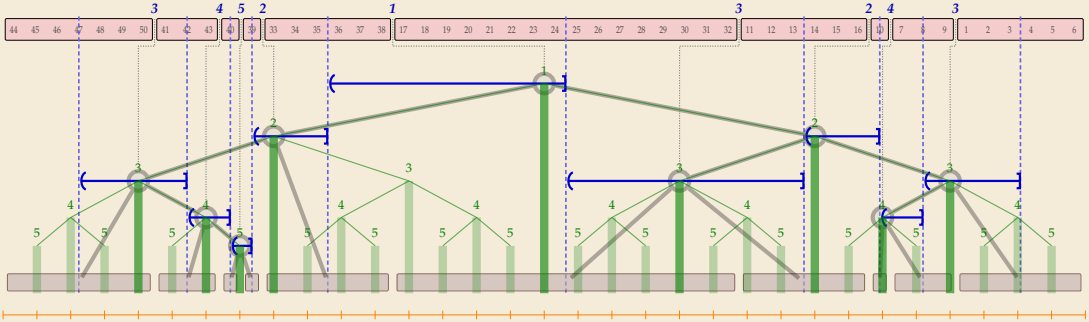
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Powersort – Run-Boundary Powers



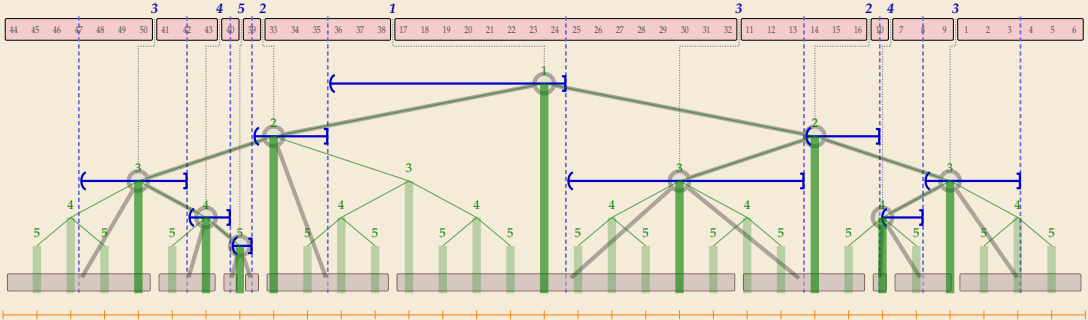
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Powersort – Run-Boundary Powers



- (virtual) perfect balanced binary tree
- midpoint intervals “snap” to closest virtual tree node
  - ⇝ assigns each run boundary a depth

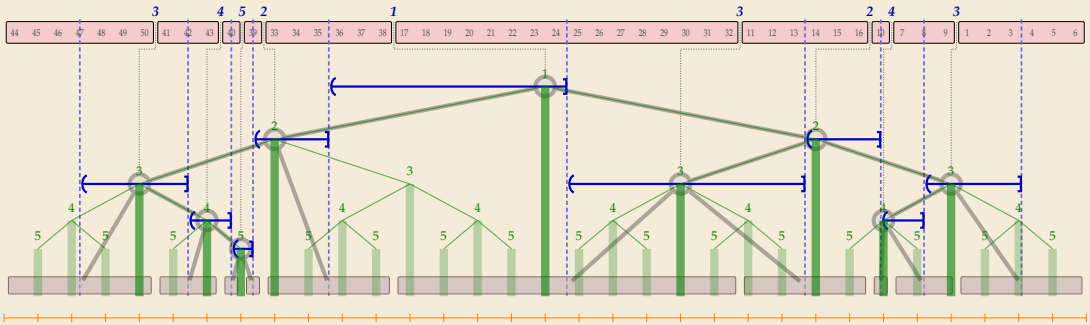
Powersort – Run-Boundary Powers



- (virtual) perfect balanced binary tree
- midpoint intervals “snap” to closest virtual tree node
  - ⇝ assigns each run boundary a depth = its *power*



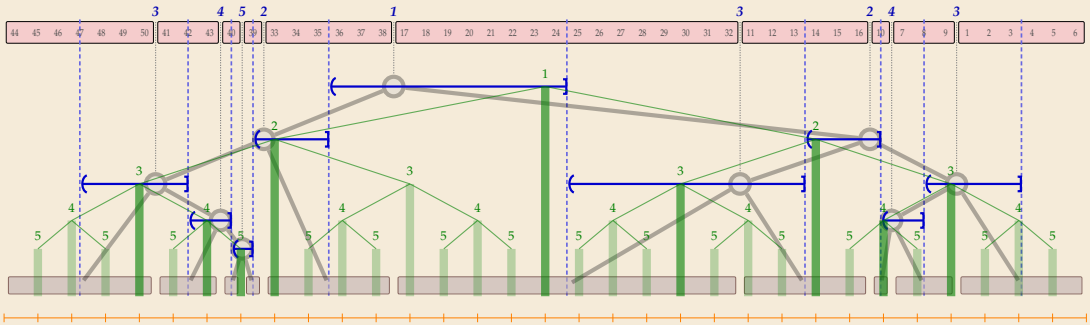
Powersort – Run-Boundary Powers



- (virtual) perfect balanced binary tree
- midpoint intervals "snap" to closest virtual tree node
  - ⇝ assigns each run boundary a depth = its *power*
- ⇝ merge tree follows virtual tree



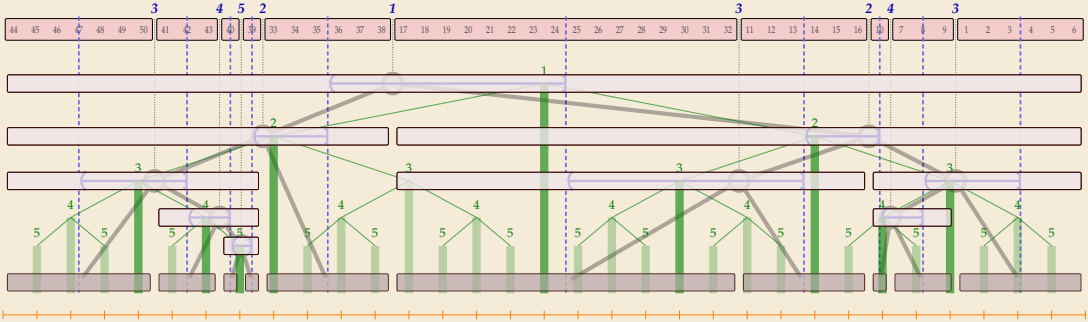
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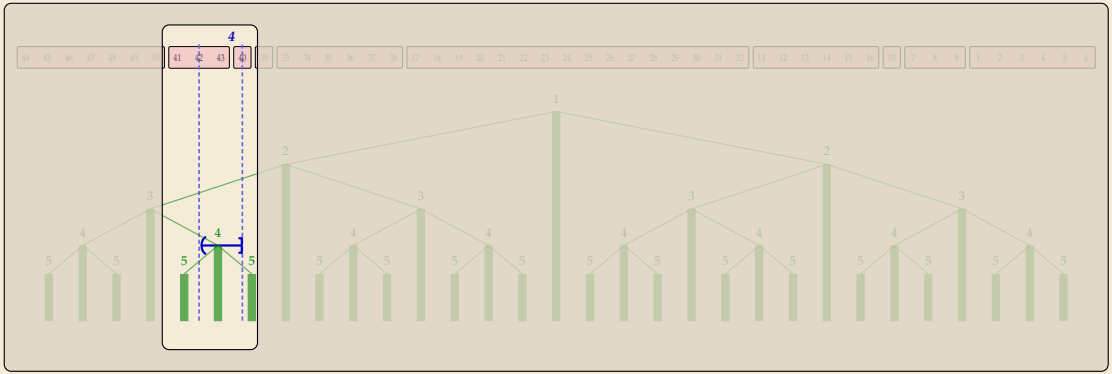
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- (virtual) perfect balanced binary tree
- midpoint intervals "snap" to closest virtual tree node
  - ⇝ assigns each run boundary a depth = its *power*
- ⇝ merge tree follows virtual tree



Powersort – Run-Boundary Powers are Local

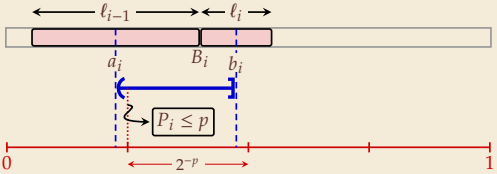


Computation of powers only depends on two adjacent runs.

Powersort – Computing powers

- Computing the power of (run boundary between) two runs
  - ⟷ = normalized midpoint interval
  - power = min ℓ s.t. ⟷ contains $c \cdot 2^{-\ell}$

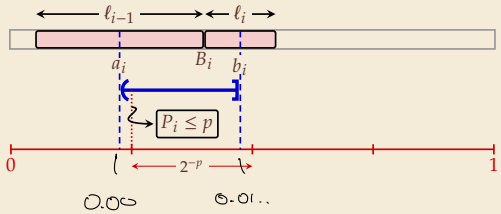




Powersort – Computing powers

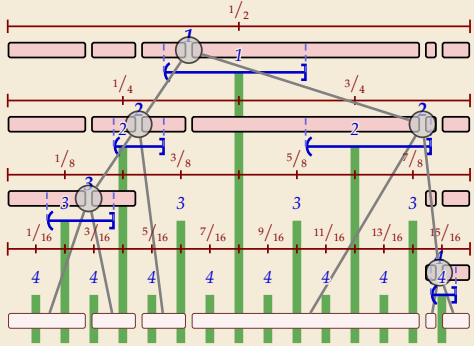
- Computing the power of (run boundary between) two runs
  - $[\!\!\;\;\!\!]$ = normalized midpoint interval
  - power = min ℓ s.t. $[\!\!\;\;\!\!]$ contains $c \cdot 2^{-\ell}$






```
1  procedure power((i1, j1), (i2, j2), n):
2      n1 := j1 − i1
3      n2 := j2 − i2
4      a := (i1 + ½n1 − 1) / n
5      b := (i2 + ½n2 − 1) / n      // interval (a, b]
6      ℓ := 0
7      while ⌊a · 2^ℓ⌋ == ⌊b · 2^ℓ⌋
8          ℓ := ℓ + 1
9      return ℓ
```

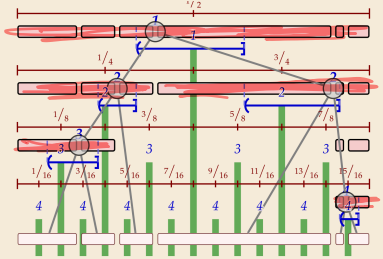
- with bitwise trickery $O(1)$ time possible



Powersort – Analysis

- **Given:** input with runs $R_1, \ldots, R_r$ of *lengths* $\ell_1, \ldots, \ell_r$;
  $n = \ell_1 + \cdots + \ell_r$

- Mergecost: $M = \sum_{r=1}^{r} \ell_i \cdot \mathrm{depth}(R_i)$



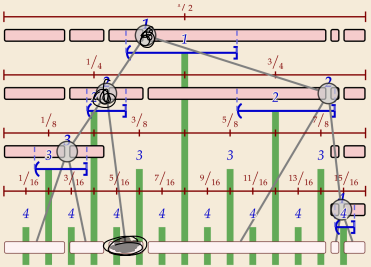
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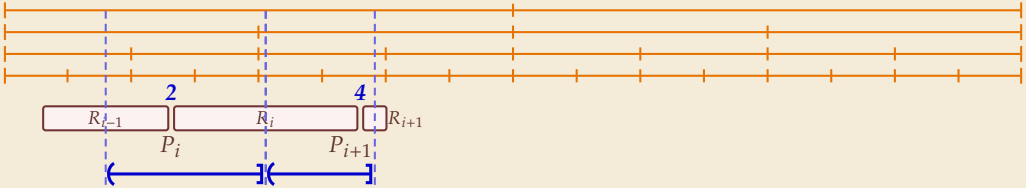
- Mergecost: $M = \sum_{r=1}^{r} \ell_i \cdot \text{depth}(R_i)$

  depth(internal node) ≤ power

  $\rightsquigarrow M \leq \sum_{r=1}^{r} \ell_i \cdot \hat{P}_i$, where $\hat{P}_i = \max\{P_i, P_{i+1}\}$



P_i = power of boundary between R_{i-1} and R_i, $i = 2, \ldots, r$
$P_1 = P_{r+1} = 0$



Powersort – Analysis

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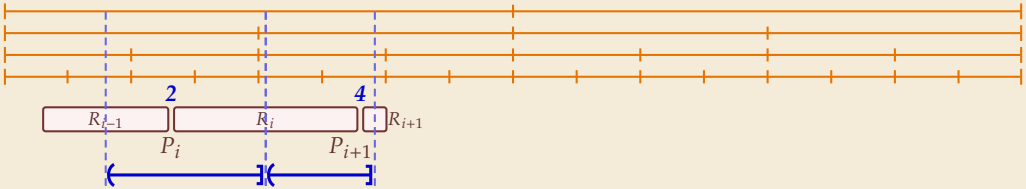
$\rightsquigarrow$ $M \leq \sum_{r=1}^{r} \ell_i \cdot \hat{P}_i$, where $\hat{P}_i = \max\{P_i, P_{i+1}\}$

$\rightsquigarrow$ To prove $M \leq n(\mathcal{H} + 2)$,
it suffices to show:

$P_i \leq \lceil \log_2(\frac{n}{\ell_i}) + 1 \rceil \leq \log_2(\frac{n}{\ell_i}) + 2$

$P_{i+1} \leq \lceil \log_2(\frac{n}{\ell_i}) + 1 \rceil \leq \log_2(\frac{n}{\ell_i}) + 2$

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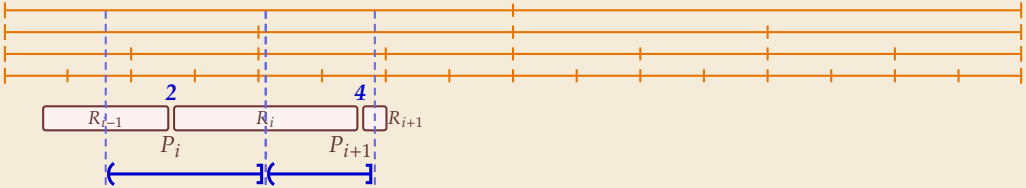


Powersort – Analysis

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- by def.: $\frac{|\longleftrightarrow|}{n} \geq 2^{-p}$ $\rightsquigarrow$ $P_i \leq p$

Powersort – Analysis

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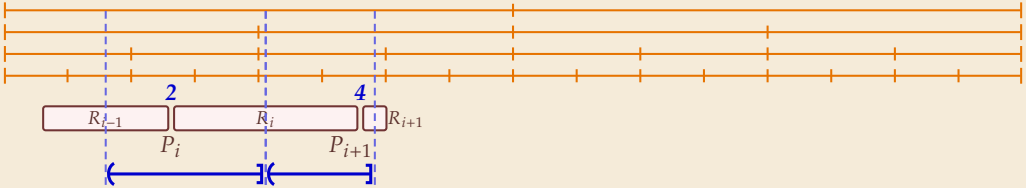
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- by def.: $\frac{|\longleftrightarrow|}{n} \geq 2^{-p} \quad\rightsquigarrow\quad P_i \leq p$
- we have: $|\longleftrightarrow| \ \geq\ \ell_i/2$

P_i = power of boundary between R_{i-1} and R_i, $i = 2, \ldots, r$
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Powersort – Analysis

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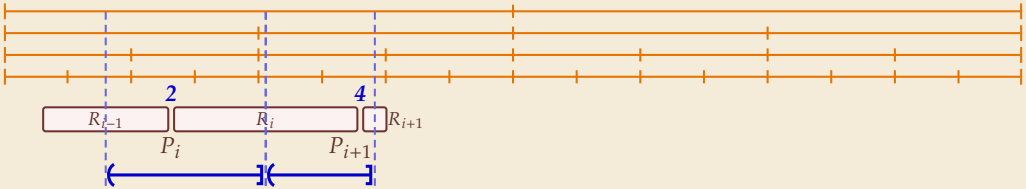
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- we have: $|\longleftrightarrow| \geq \ell_i/2$

$\rightsquigarrow$ $P_i \leq \lceil \log_2(\frac{n}{\ell_i}) \rceil + 1$ ■



Powersort – Discussion



Retains all advantages of Timsort

- good locality in memory accesses
- no recursion
- all the tricks in Timsort



optimally adapts to existing runs



minimal overhead for finding merge order